



Slicer Developer Tutorial: Programming in Slicer

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Goal of the tutorial



```
def threshold(t):  
    n=getNode('T2')  
    a=array('T2')  
    a[a<t]=0  
    arrayFromVolumeModified(n)  
    print('Thresholding done')
```



```
b=qt.QPushButton('Toggle')  
b.connect('clicked()',toggle)  
b.styleSheet = "font-size: 24pt; color:  
aqua; margin: 20px"  
b.show()
```

This tutorial is an introduction to the Python interactor and the Qt widget toolkit in 3D Slicer release version 5

Tutorial Outline



Part 1: 3D Slicer Modules Overview



Part 2: Getting Familiar with the Python environment in 3D Slicer

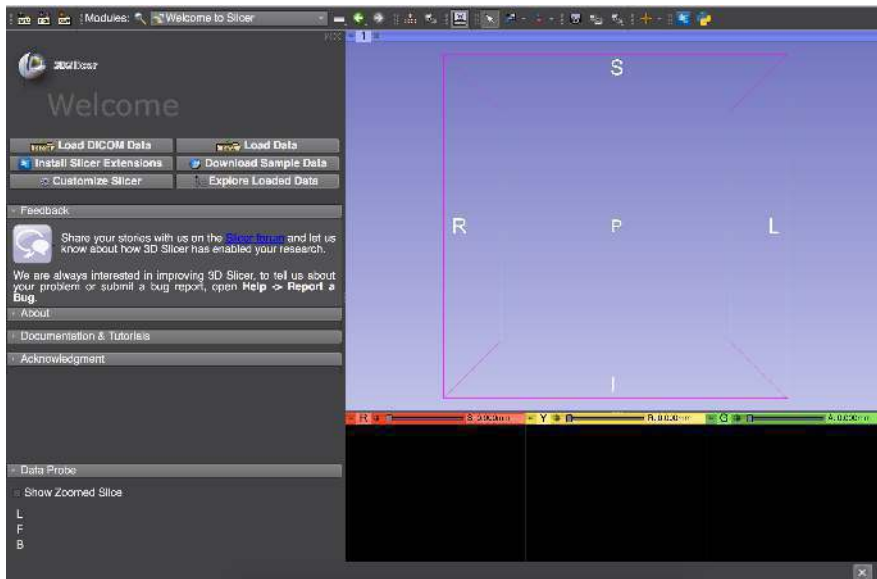


Part 3: Getting Familiar with the Qt widget toolkit in 3D Slicer

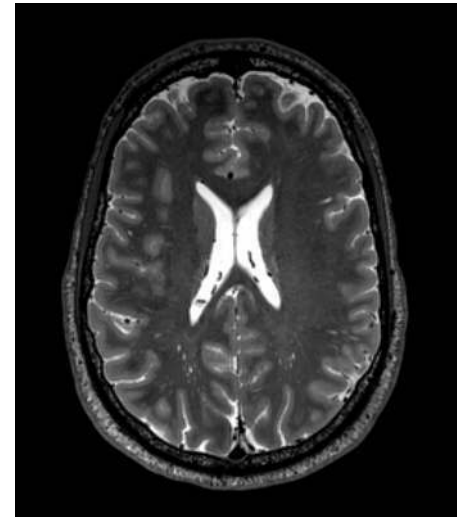
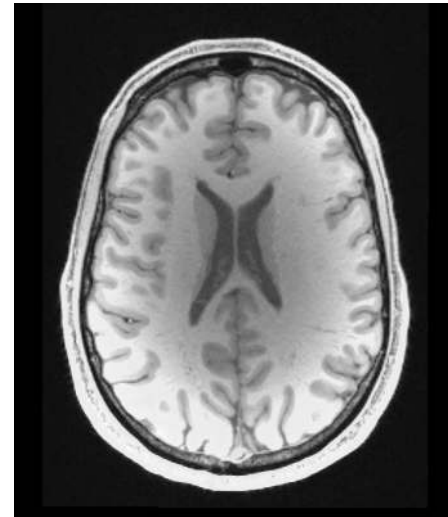
Disclaimer

- 3D Slicer is a free open source software application distributed under a BSD style license.
- The software is not FDA approved or CE-Marked, and is for research use only.

Tutorial materials



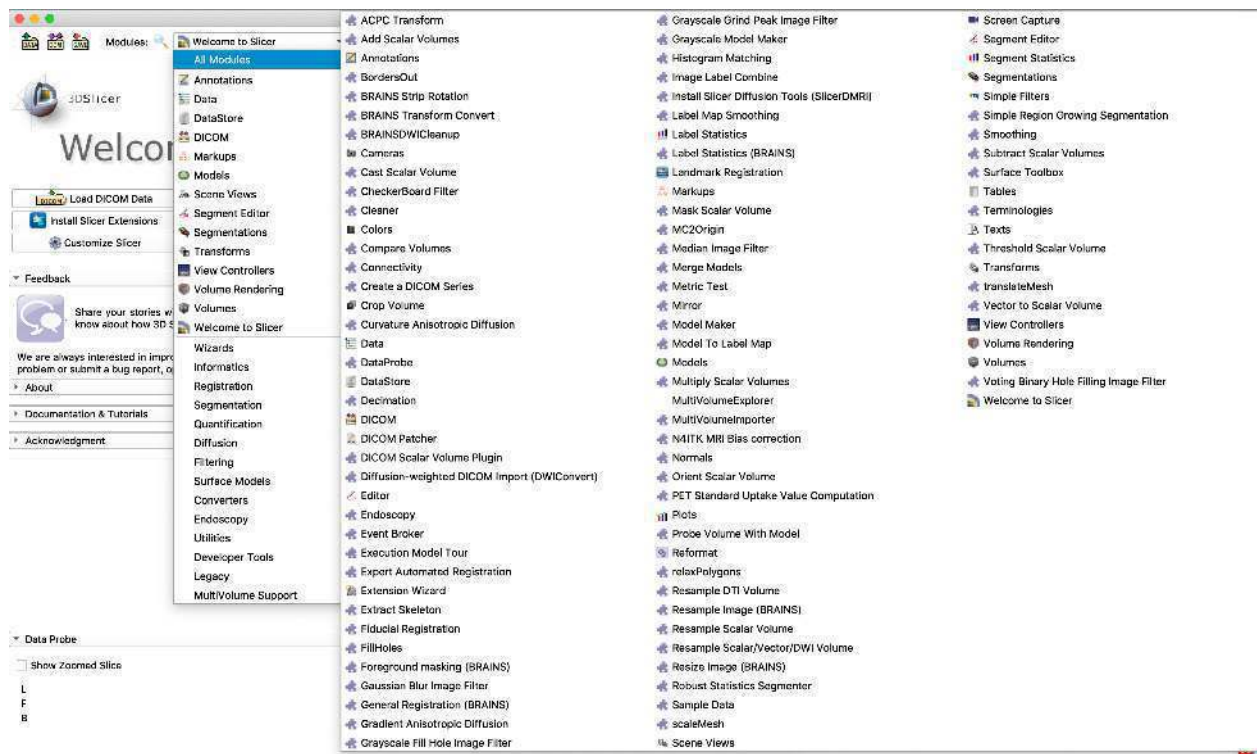
3D Slicer version 4.11



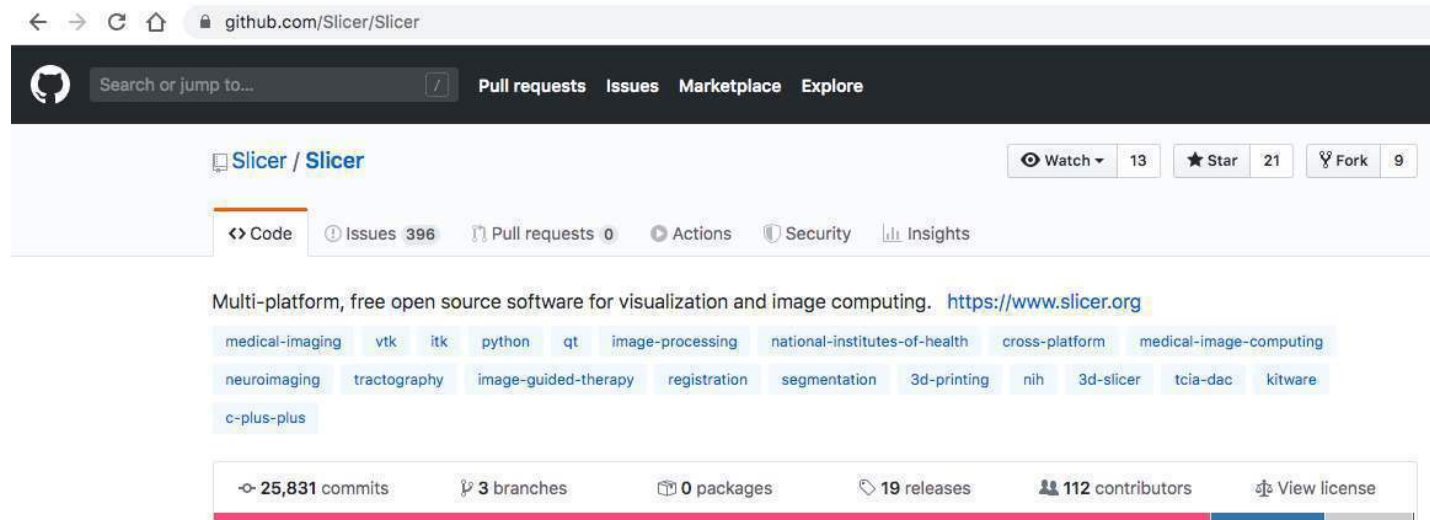
SlicerProgrammingTutorialData.zip

Part 1

Slicer Modules Overview



3D Slicer



- 3D Slicer is an open-source platform for the analysis and visualization of medical imaging data
- 3D Slicer is compiled and tested every day on Windows, Mac, and Linux platforms
- The source code is freely available on GitHub at <http://github.com/Slicer/Slicer>

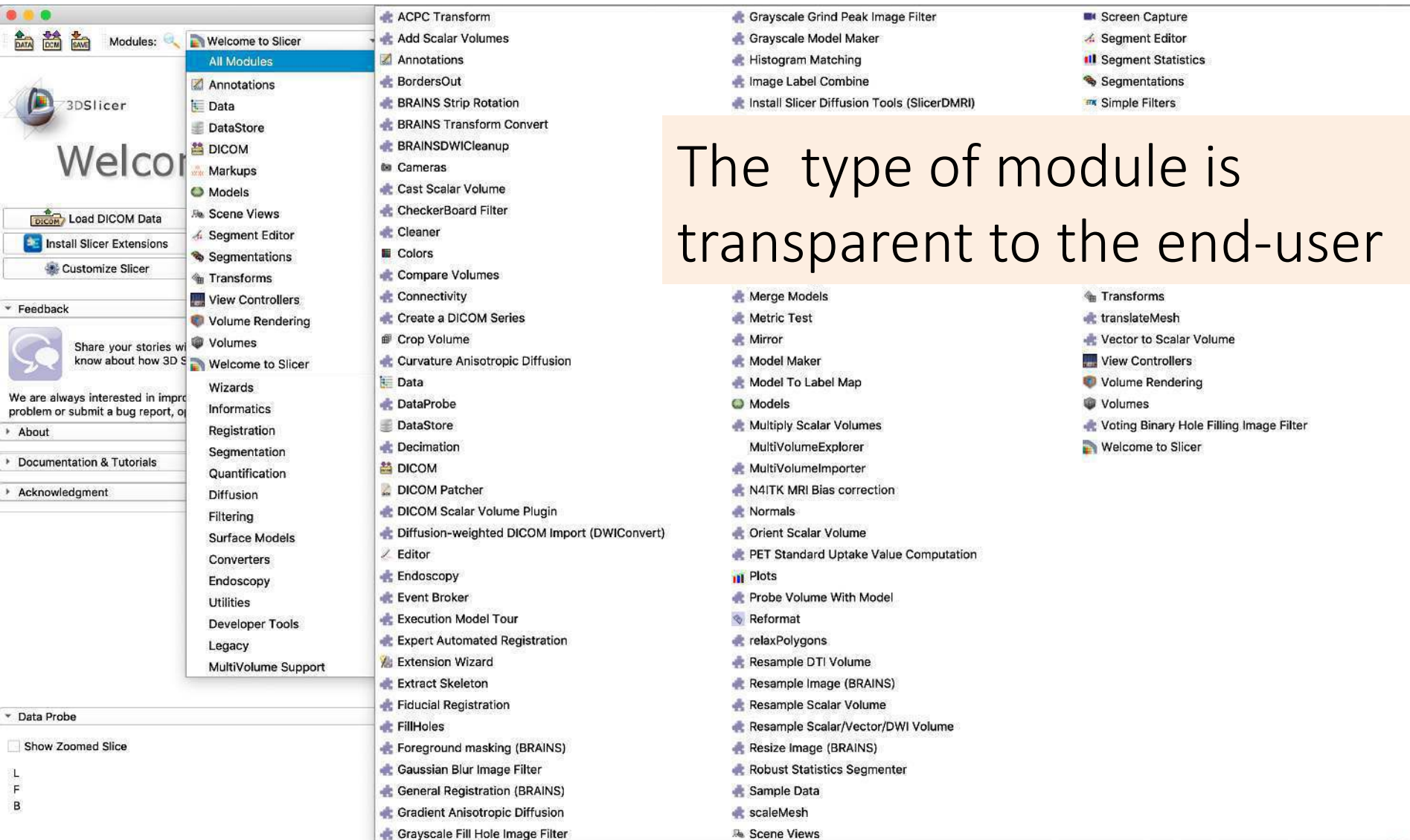
Slicer Modules

3D Slicer supports three types of modules:

- **Command Line Interface (CLI):** standalone executable with limited input/output arguments
- **Loadable Modules (C++ Plugins):** optimized for heavy computation
- **Scripted Modules (Python):** recommended for fast prototyping and workflow development

Focus of this tutorial

Slicer Modules



The screenshot displays the Slicer application's 'Modules' menu, which lists various modules available for use. The menu is organized into a hierarchical structure, with 'All Modules' selected. The modules are listed in three columns, each preceded by a small icon representing the module's category.

Modules:

- Welcome to Slicer
- All Modules
- Annotations
- Data
- DataStore
- DICOM
- Markups
- Models
- Scene Views
- Segment Editor
- Segmentations
- Transforms
- View Controllers
- Volume Rendering
- Volumes
- Welcome to Slicer
- Wizards
- Informatics
- Registration
- Segmentation
- Quantification
- Diffusion
- Filtering
- Surface Models
- Converters
- Endoscopy
- Utilities
- Developer Tools
- Legacy
- MultiVolume Support

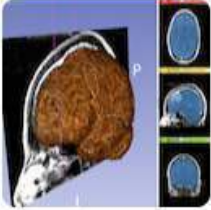
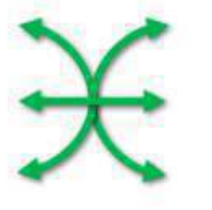
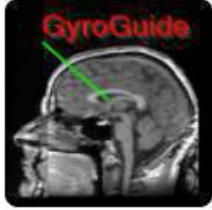
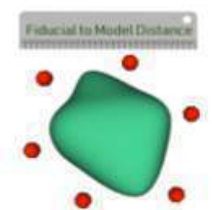
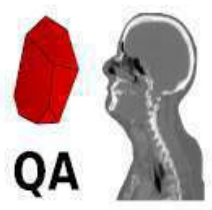
Available Modules:

- ACPC Transform
- Add Scalar Volumes
- Annotations
- BordersOut
- BRAINS Strip Rotation
- BRAINS Transform Convert
- BRAINS DWICleanup
- Cameras
- Cast Scalar Volume
- CheckerBoard Filter
- Cleaner
- Colors
- Compare Volumes
- Connectivity
- Create a DICOM Series
- Crop Volume
- Curvature Anisotropic Diffusion
- Data
- DataProbe
- DataStore
- Decimation
- DICOM
- DICOM Patcher
- DICOM Scalar Volume Plugin
- Diffusion-weighted DICOM Import (DWIConvert)
- Editor
- Endoscopy
- Event Broker
- Execution Model Tour
- Expert Automated Registration
- Extension Wizard
- Extract Skeleton
- Fiducial Registration
- FillHoles
- Foreground masking (BRAINS)
- Gaussian Blur Image Filter
- General Registration (BRAINS)
- Gradient Anisotropic Diffusion
- Grayscale Fill Hole Image Filter
- Grayscale Grind Peak Image Filter
- Grayscale Model Maker
- Histogram Matching
- Image Label Combine
- Install Slicer Diffusion Tools (SlicerDMRI)
- Merge Models
- Metric Test
- Mirror
- Model Maker
- Model To Label Map
- Models
- Multiply Scalar Volumes
- MultiVolumeExplorer
- MultiVolumeImporter
- N4ITK MRI Bias correction
- Normals
- Orient Scalar Volume
- PET Standard Uptake Value Computation
- Plots
- Probe Volume With Model
- Reformat
- relaxPolygons
- Resample DTI Volume
- Resample Image (BRAINS)
- Resample Scalar Volume
- Resample Scalar/Vector/DWI Volume
- Resize Image (BRAINS)
- Robust Statistics Segmenter
- Sample Data
- scaleMesh
- Scene Views
- Screen Capture
- Segment Editor
- Segment Statistics
- Segmentations
- Simple Filters
- Transforms
- translateMesh
- Vector to Scalar Volume
- View Controllers
- Volume Rendering
- Volumes
- Voting Binary Hole Filling Image Filter
- Welcome to Slicer

The type of module is transparent to the end-user

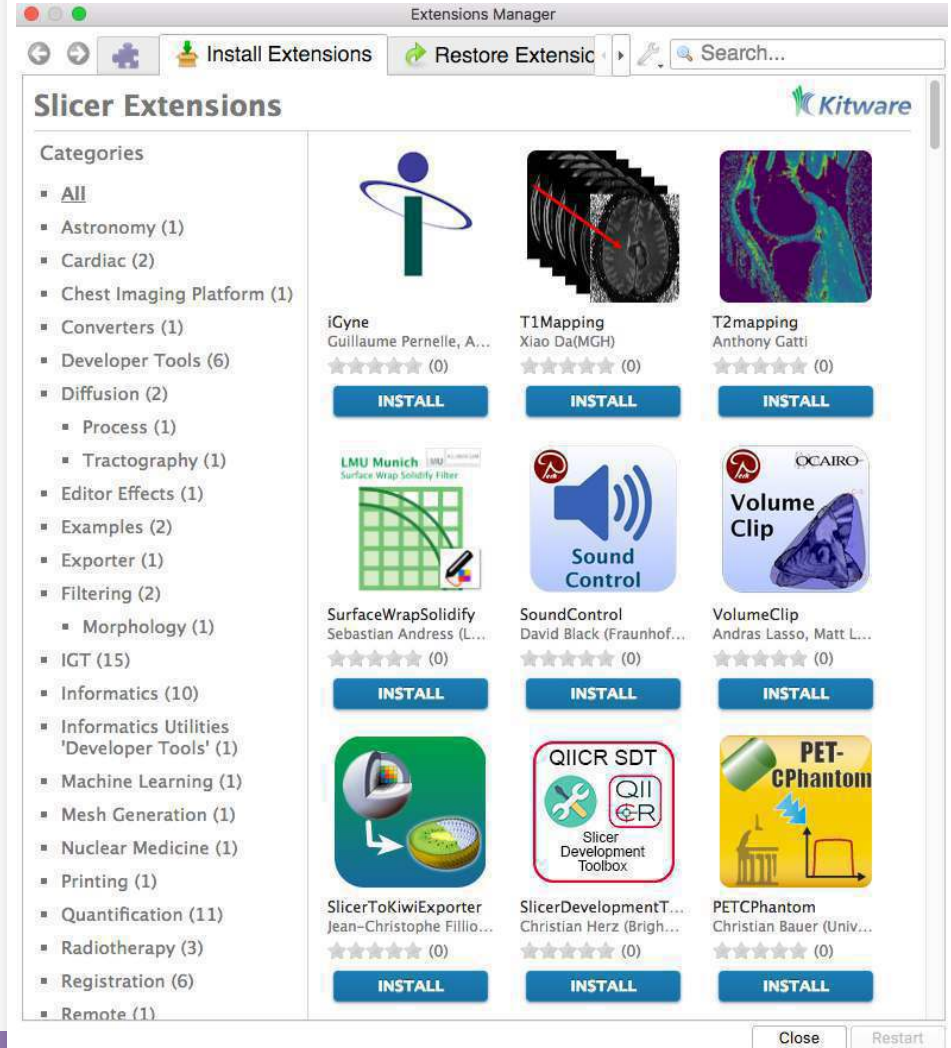
Slicer Extensions

A Slicer Extension is a delivery package bundling together one or more Slicer modules

			
SwissSkullStripper Bill Lorensen (Noware...) ★★★★★ (0)	PETTumorSegmenta... Christian Bauer (Univ...) ★★★★★ (0)	SlicerOpenIGTLink Junichi Tokuda (SPL), ... ★★★★★ (0)	GyroGuide Ruifeng Chen, Luping... ★★★★★ (0)
INSTALL	INSTALL	INSTALL	INSTALL
			
PET-IndiC Ethan Ulrich (Universi...) ★★★★★ (0)	FiducialsToModelDi... Jesse Reynolds (Cante...) ★★★★★ (0)	Slicer-Wasp Thomas Lawson (MR...) ★★★★★ (0)	ImageCompare Paolo Zaffino (Magna ...) ★★★★★ (0)
INSTALL	INSTALL	INSTALL	INSTALL

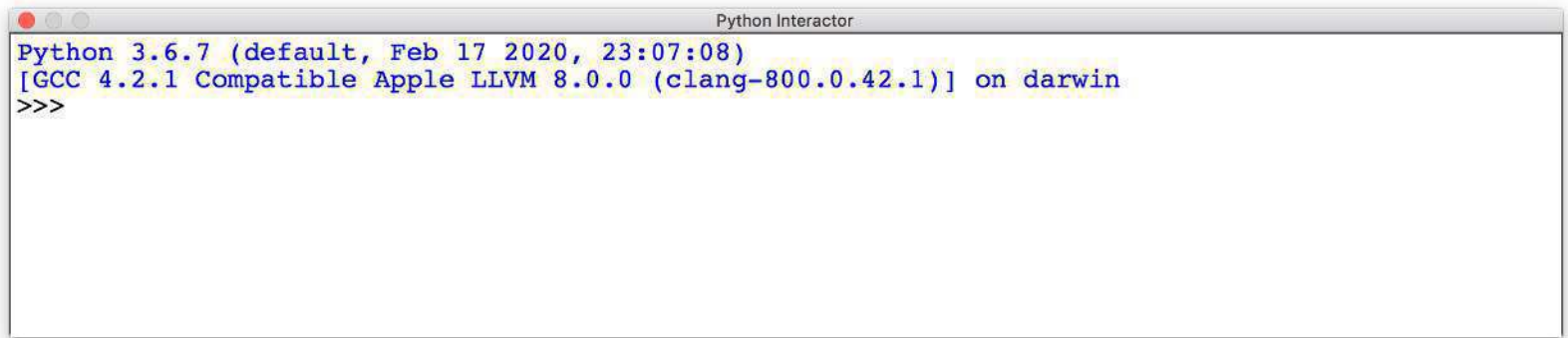
Slicer Extension Manager

- The Slicer Extension Manager provides an 'App store' platform for the 3D Slicer ecosystem
- The Extension Manager enables an easy creation and installation of Slicer extensions
- Slicer release version 5 includes over 130 extensions



Part 2

Getting Familiar with the Python environment in 3D Slicer



```
Python Interactor
Python 3.6.7 (default, Feb 17 2020, 23:07:08)
[GCC 4.2.1 Compatible Apple LLVM 8.0.0 (clang-800.0.42.1)] on darwin
>>>
```

Python in Slicer

Slicer v.4.11 works with **Python3** and a rich set of standard libraries



NumPy is the fundamental package for scientific computing with Python.



VTK is an open-source library for manipulating and displaying scientific data.



ITK is an open-source library for image analysis.

CTK is an open-source library for biomedical image computing.



PythonQt is a Python binding for Qt.



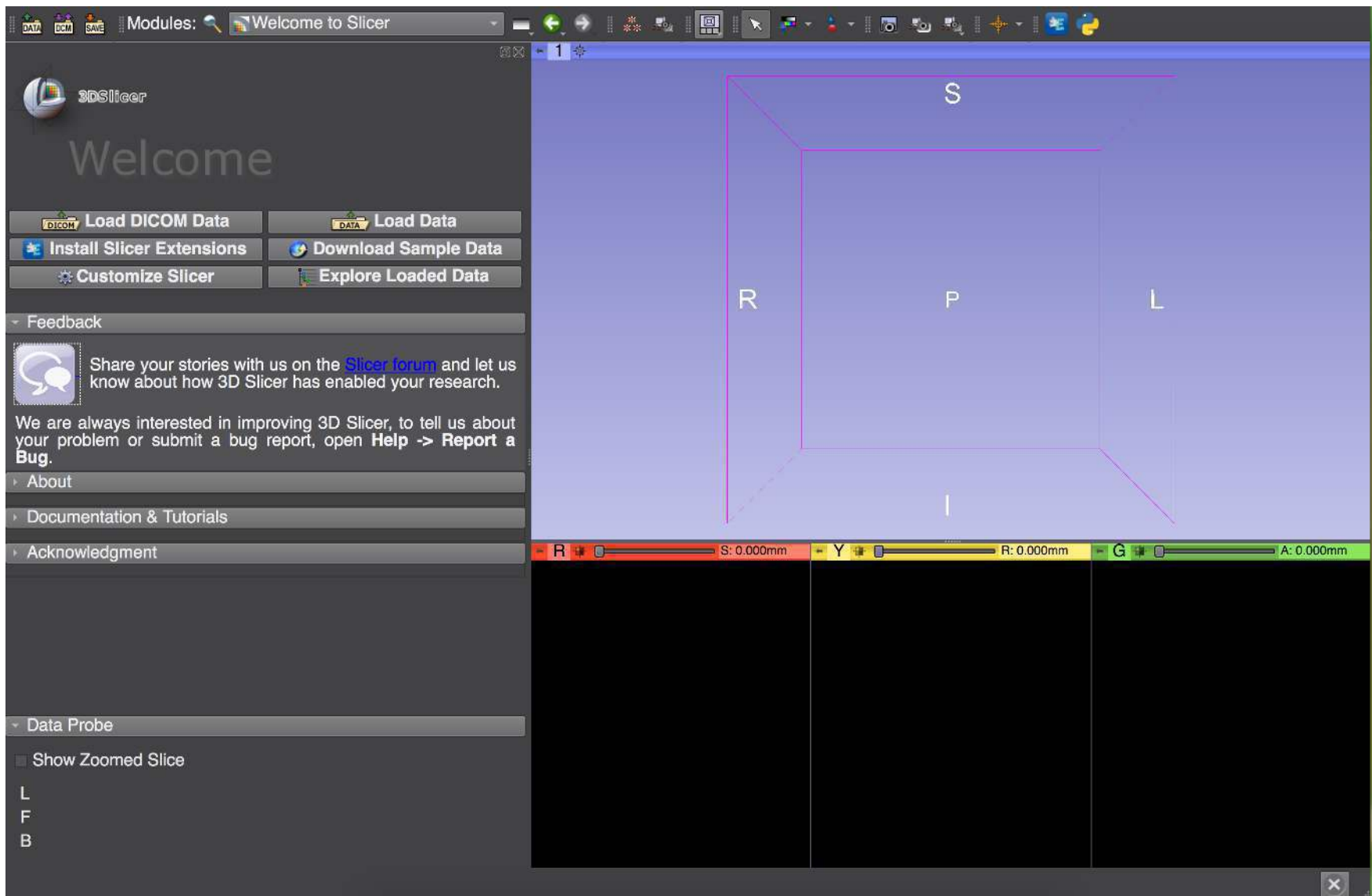
Qt is a cross-platform framework used as a graphical toolkit.

Python in Slicer



The **Python Package index (PyPi)** gives access to over 200,000 additional Python packages (<http://pipy.org>)

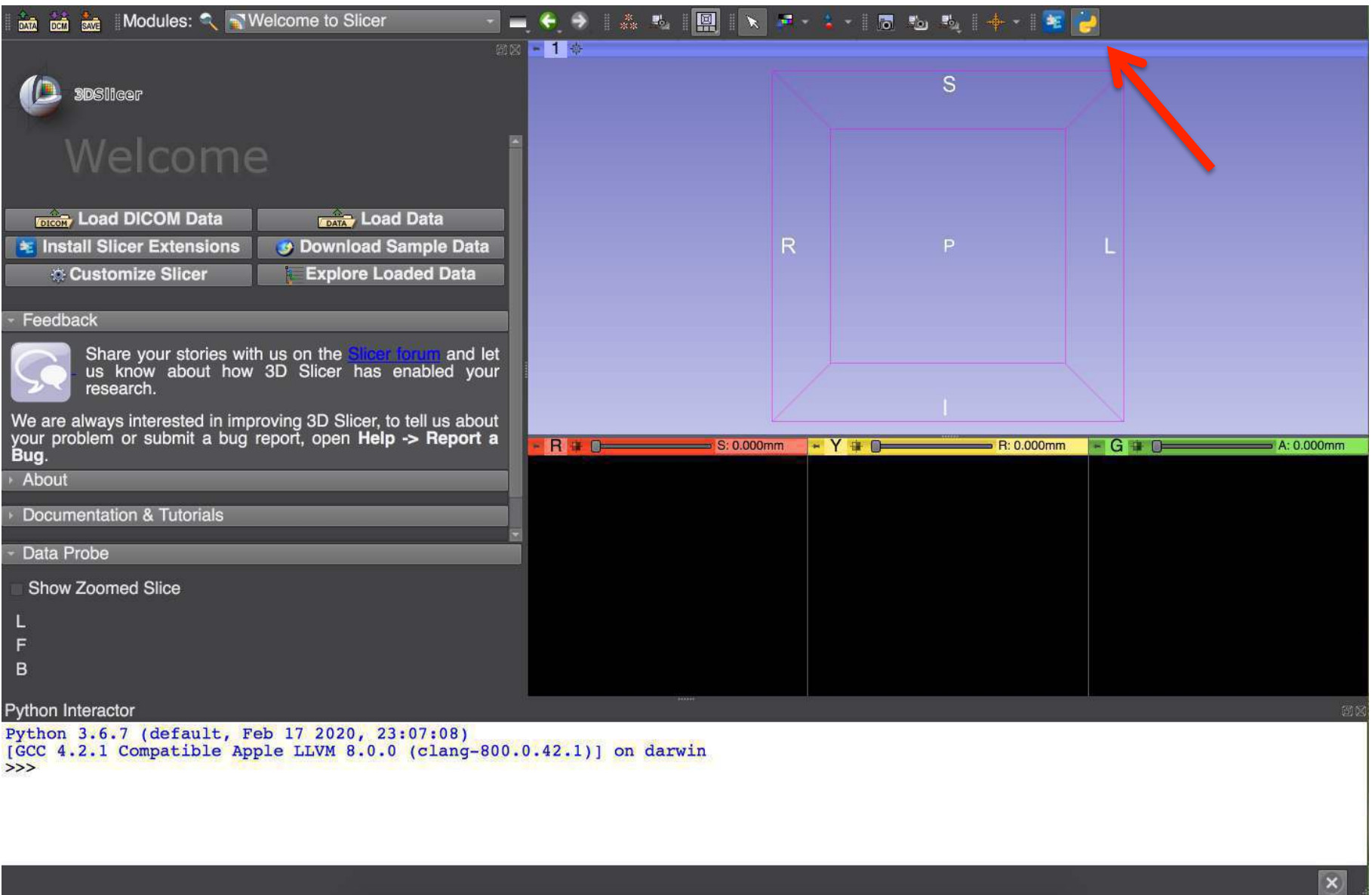
- The **pip install** command in Slicer enables developers to install most common scientific computing tools (e.g. TensorFlow, SciPy, PyTorch, Pandas, etc.)
- Slicer can be used as a **Jupyter notebook** kernel
- PyCharm and other Python development tools can be used with Slicer



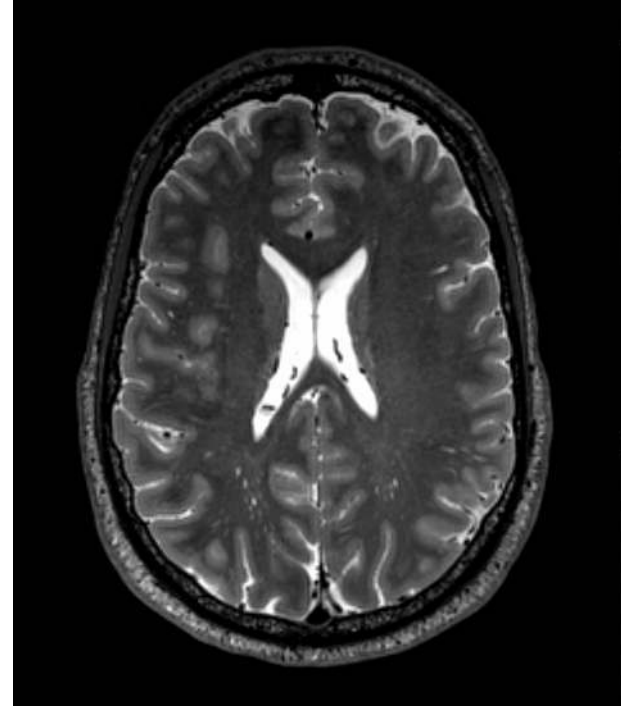
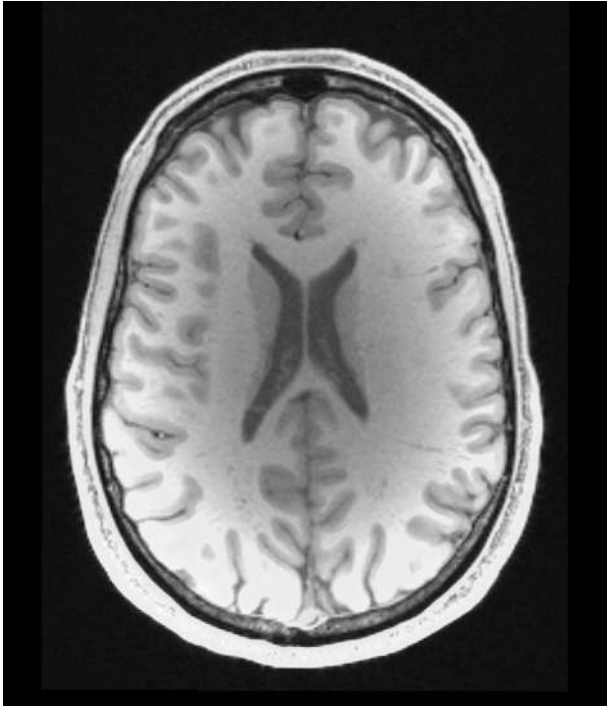
Slicer release version 5 integrates
Python3, VTK5 and ITK5

The Python Console in Slicer

The Python Interactor is a Qt-based console that enables direct access to Slicer MRML Nodes, libraries (NumPy, VTK, ITK, CTK) and Qt.

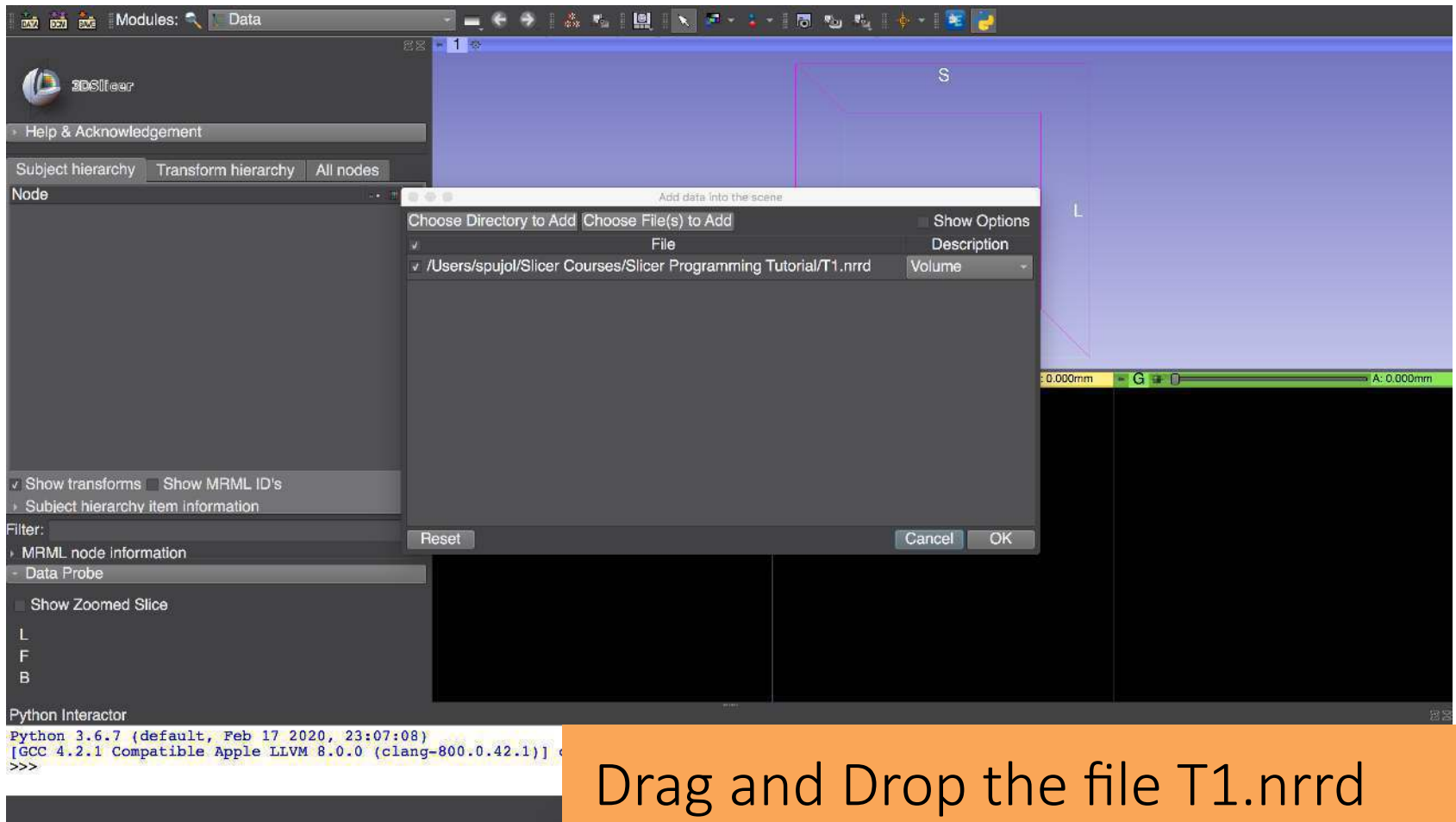


To access the Python Interactor, click on the Python icon  in the top bar menu of Slicer.



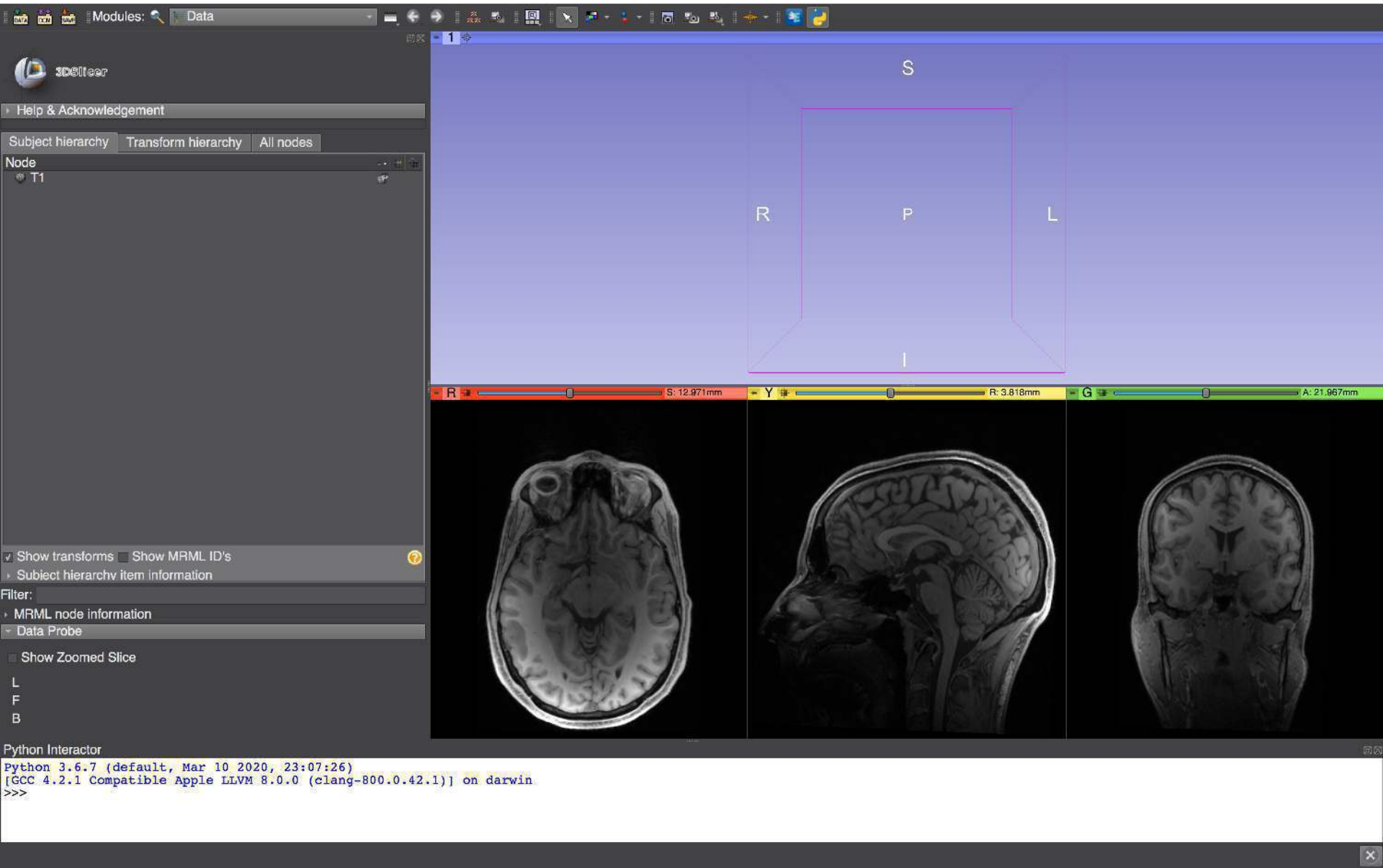
The Slicer Programming tutorial dataset includes a T1-weighted and a T2-weighted MRI scan of a healthy subject

Tutorial dataset



Drag and Drop the file T1.nrrd
Click on OK to load the file in Slicer

Tutorial dataset



Big Picture

- Slicer is free and open-source software
- There are thousands of sophisticated medical images available on the Internet that you could visualize and analyze with 3D Slicer

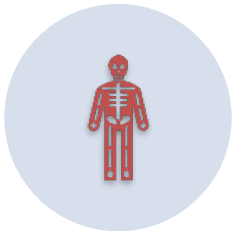
Slicer Data Model



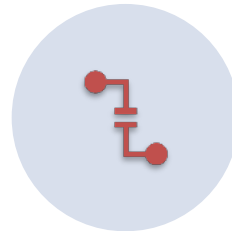
The **Slicer Data Model** is based on the Slicer Scene Data Structure



A **Slicer scene** is a collection of images, annotations, 3D models, spatial transforms, fiducials and cameras



The **Medical Reality Markup Language (MRML)** is an XML-based language used to serialize the content of Slicer scene on disk (scene.mrml)



Each element a scene is called a **MRML node**

Slicer MRML Nodes: Basic Types



Data Node: Stores the raw data

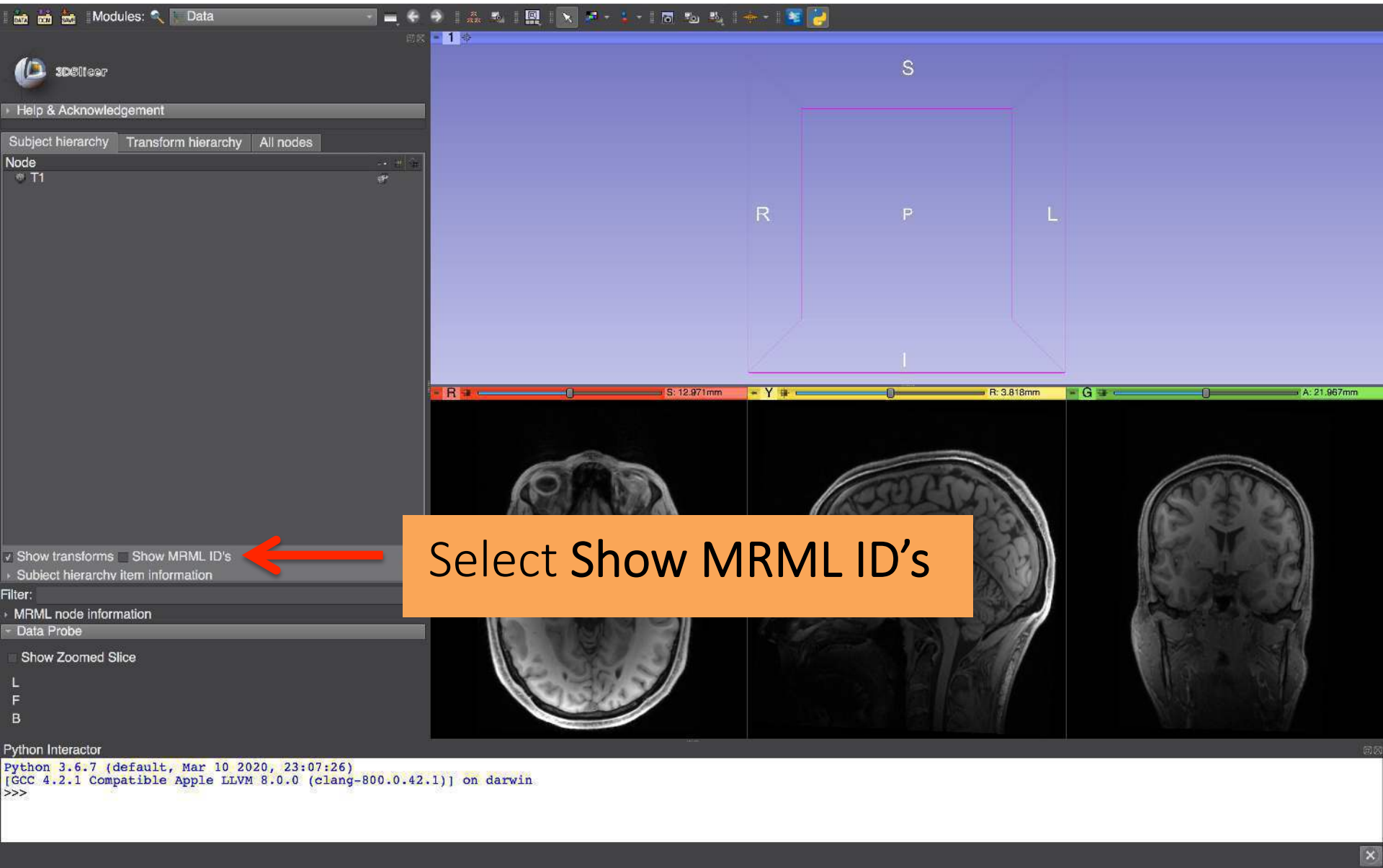


Display Node: Describes how the data should be visualized

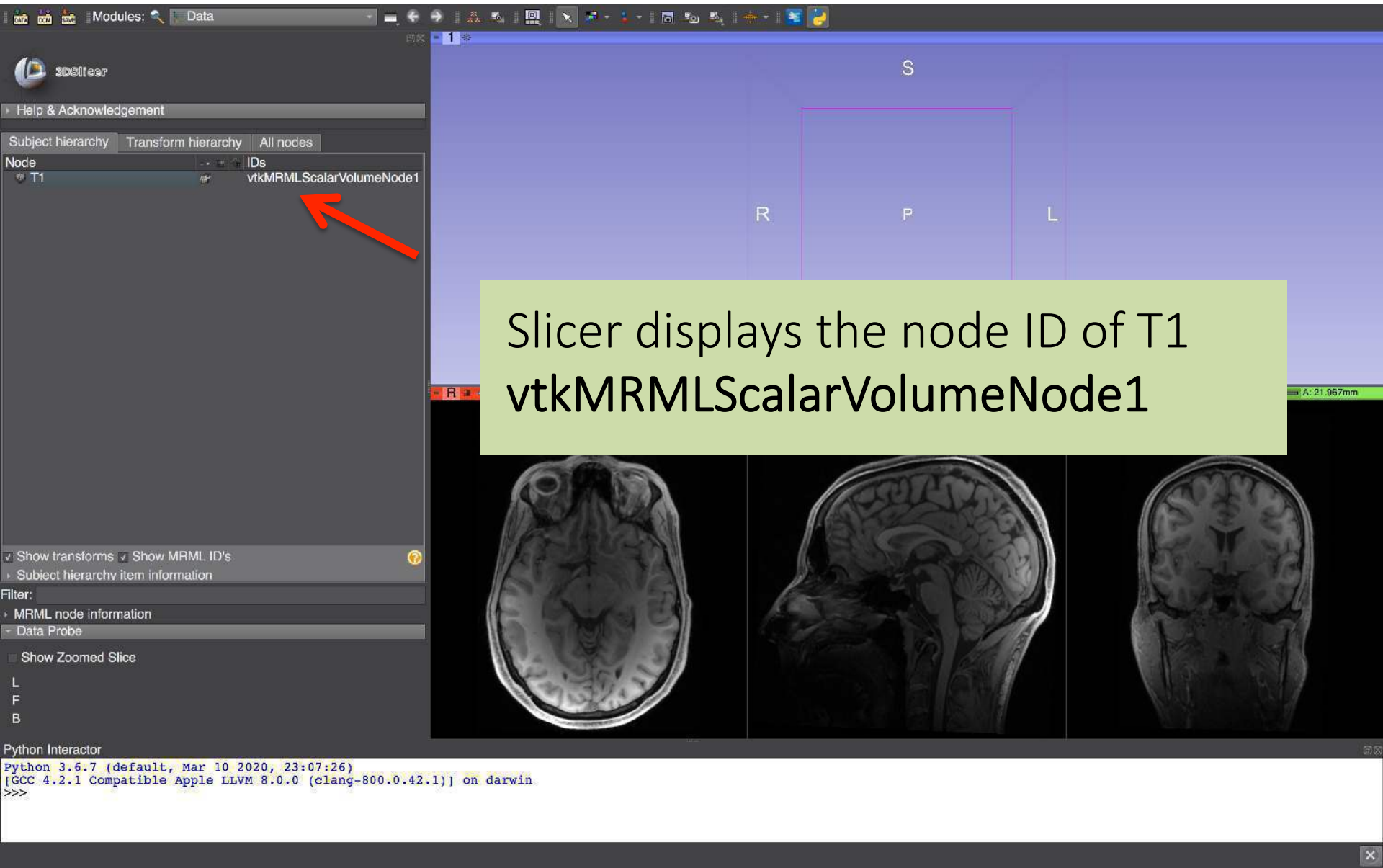


Storage Node: Describes how the data should be stored on disk

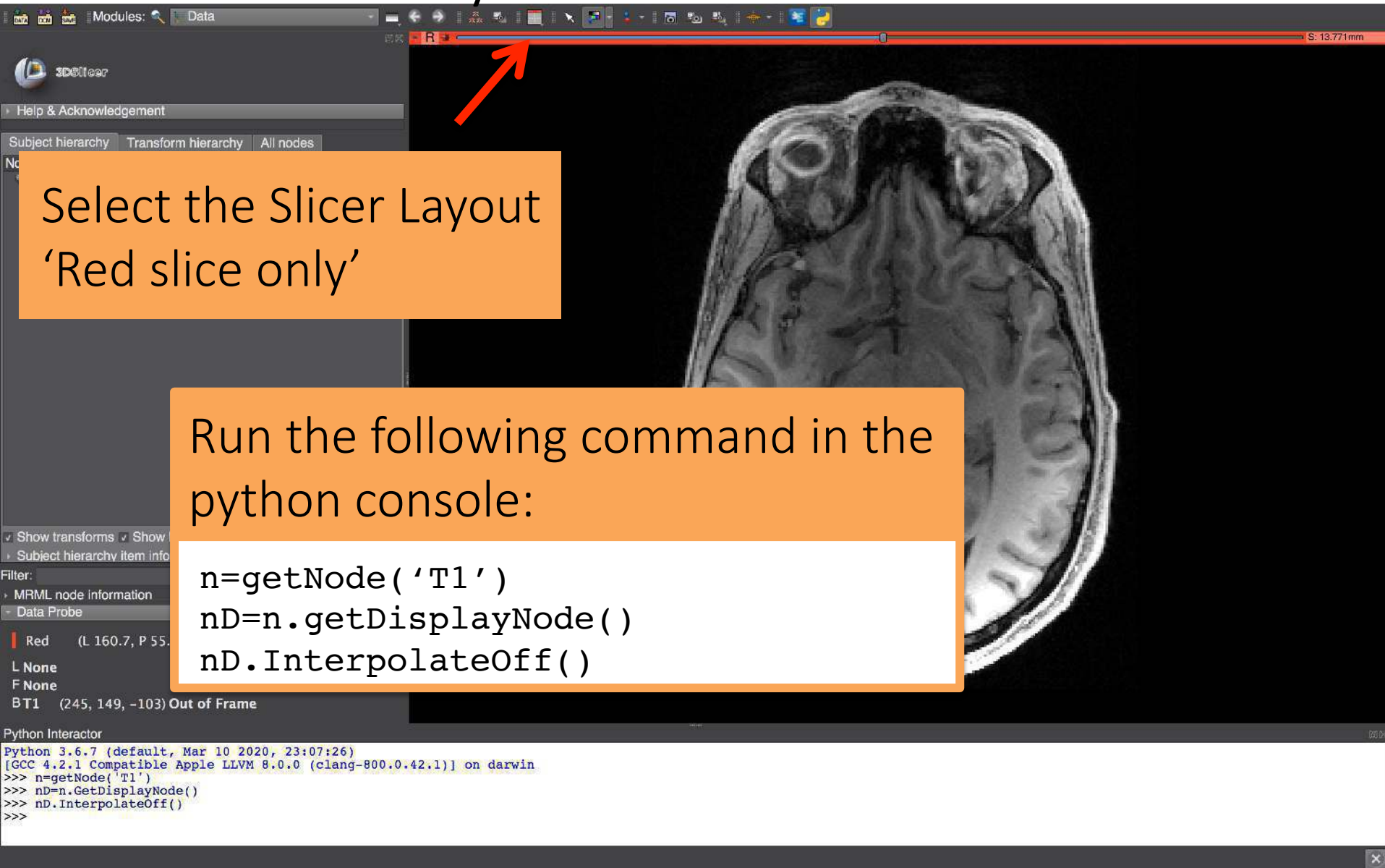
Tutorial dataset



Slicer Data Model



Accessing MRML nodes from the Python interactor



Select the Slicer Layout 'Red slice only'

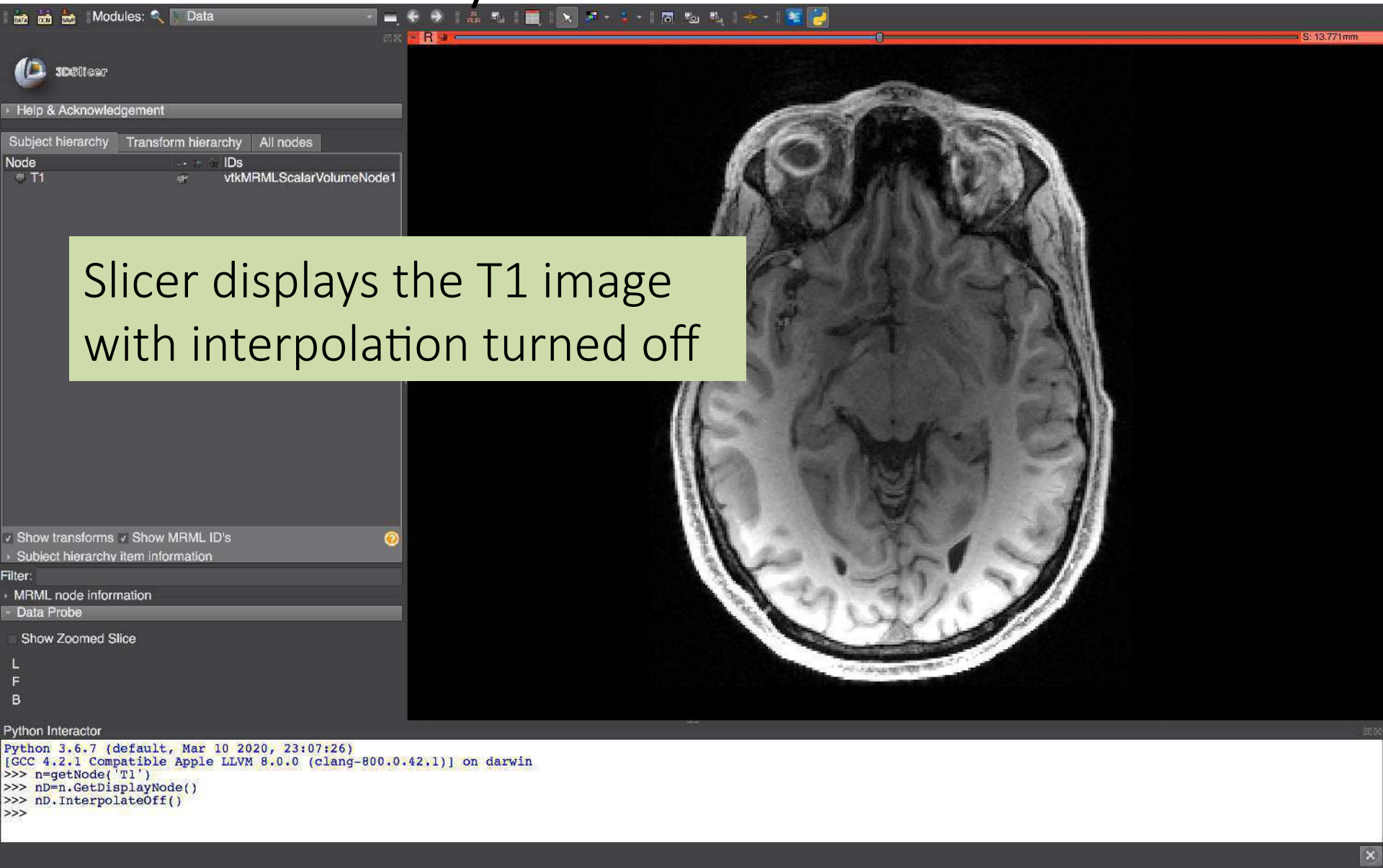
Run the following command in the python console:

```
n=getNode('T1')
nD=n.GetDisplayNode()
nD.InterpolateOff()
```

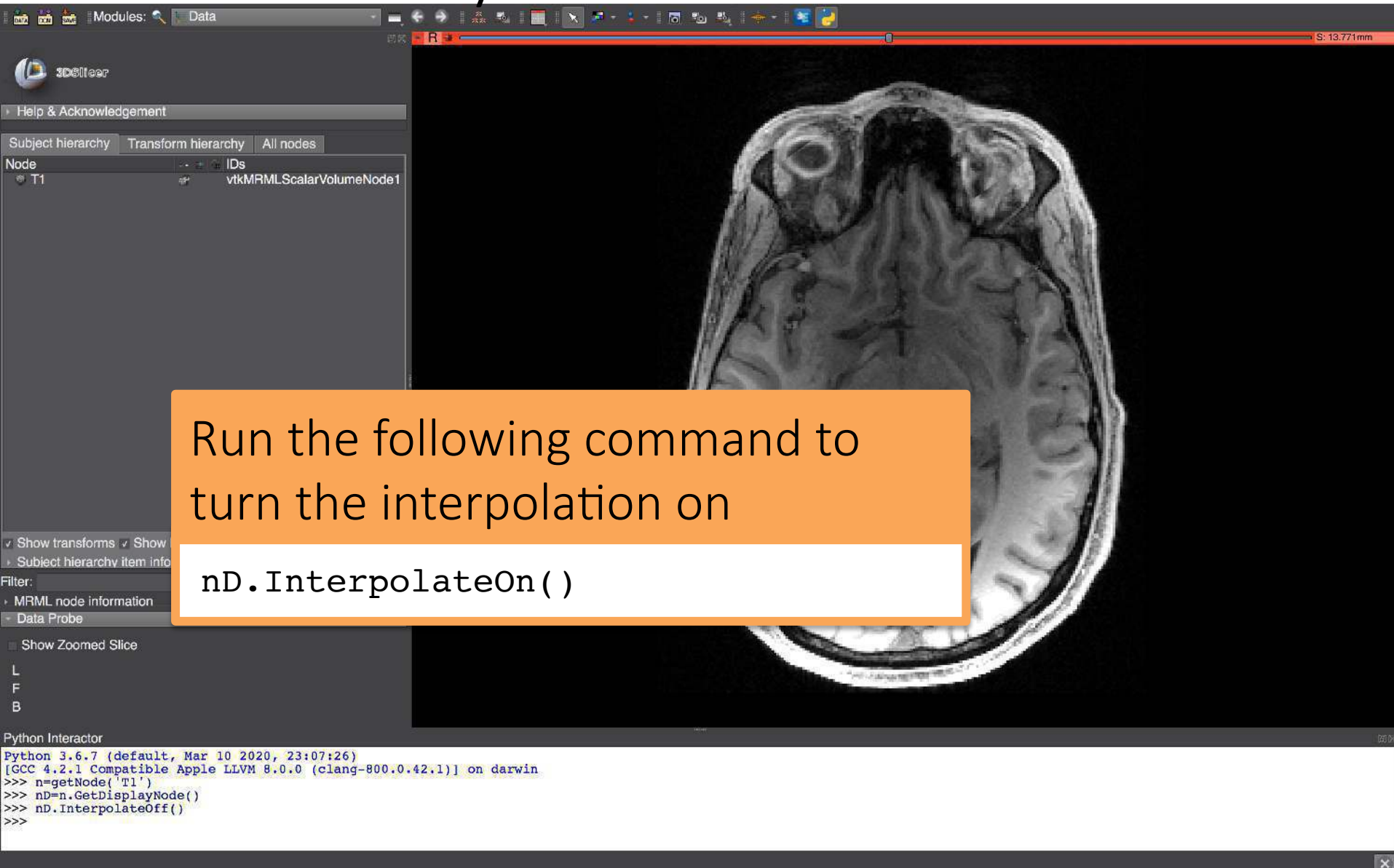
Python Interactor

```
Python 3.6.7 (default, Mar 10 2020, 23:07:26)
[GCC 4.2.1 Compatible Apple LLVM 8.0.0 (clang-800.0.42.1)] on darwin
>>> n=getNode('T1')
>>> nD=n.GetDisplayNode()
>>> nD.InterpolateOff()
>>>
```

Accessing MRML nodes from the Python interactor



Accessing MRML nodes from the Python interactor



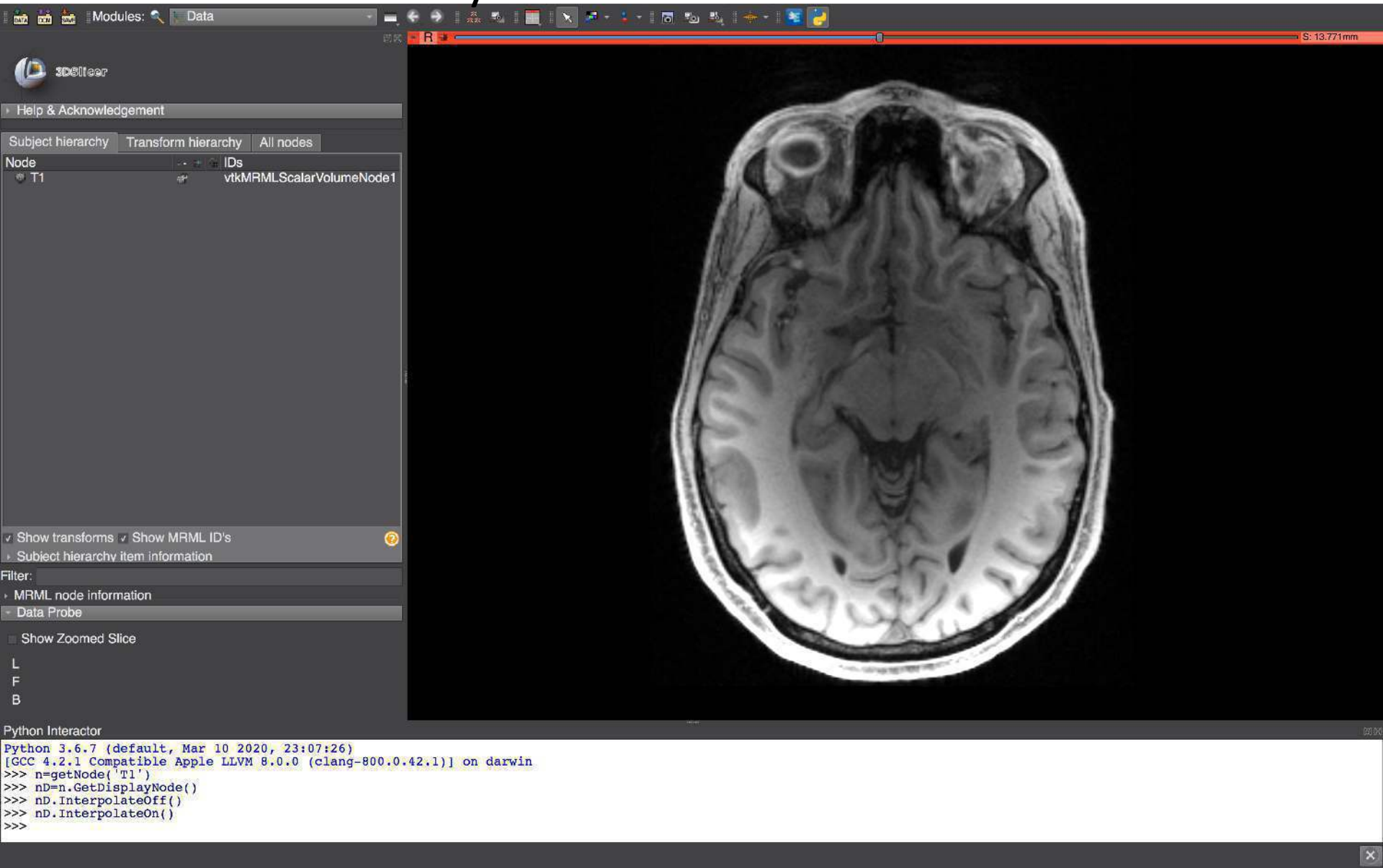
Run the following command to turn the interpolation on

```
nD.InterpolateOn()
```

Python Interactor

```
Python 3.6.7 (default, Mar 10 2020, 23:07:26)
[GCC 4.2.1 Compatible Apple LLVM 8.0.0 (clang-800.0.42.1)] on darwin
>>> n=getNode('T1')
>>> nD=n.GetDisplayNode()
>>> nD.InterpolateOff()
>>>
```

Accessing MRML nodes from the Python interactor



The screenshot displays the 3D Slicer software interface. The main window shows an axial brain MRI slice. The left sidebar contains the 'Subject hierarchy' panel, which lists the loaded data. The 'Python Interactor' panel at the bottom shows the following code and output:

```
Python Interactor
Python 3.6.7 (default, Mar 10 2020, 23:07:26)
[GCC 4.2.1 Compatible Apple LLVM 8.0.0 (clang-800.0.42.1)] on darwin
>>> n=getNode('T1')
>>> nd=n.GetDisplayNode()
>>> nd.InterpolateOff()
>>> nd.InterpolateOn()
>>>
```

The 'Subject hierarchy' panel shows the following structure:

- Node
 - T1
 - vtkMRMLScalarVolumeNode1

The 'Filter' section shows the following options:

- MRML node information
 - Data Probe

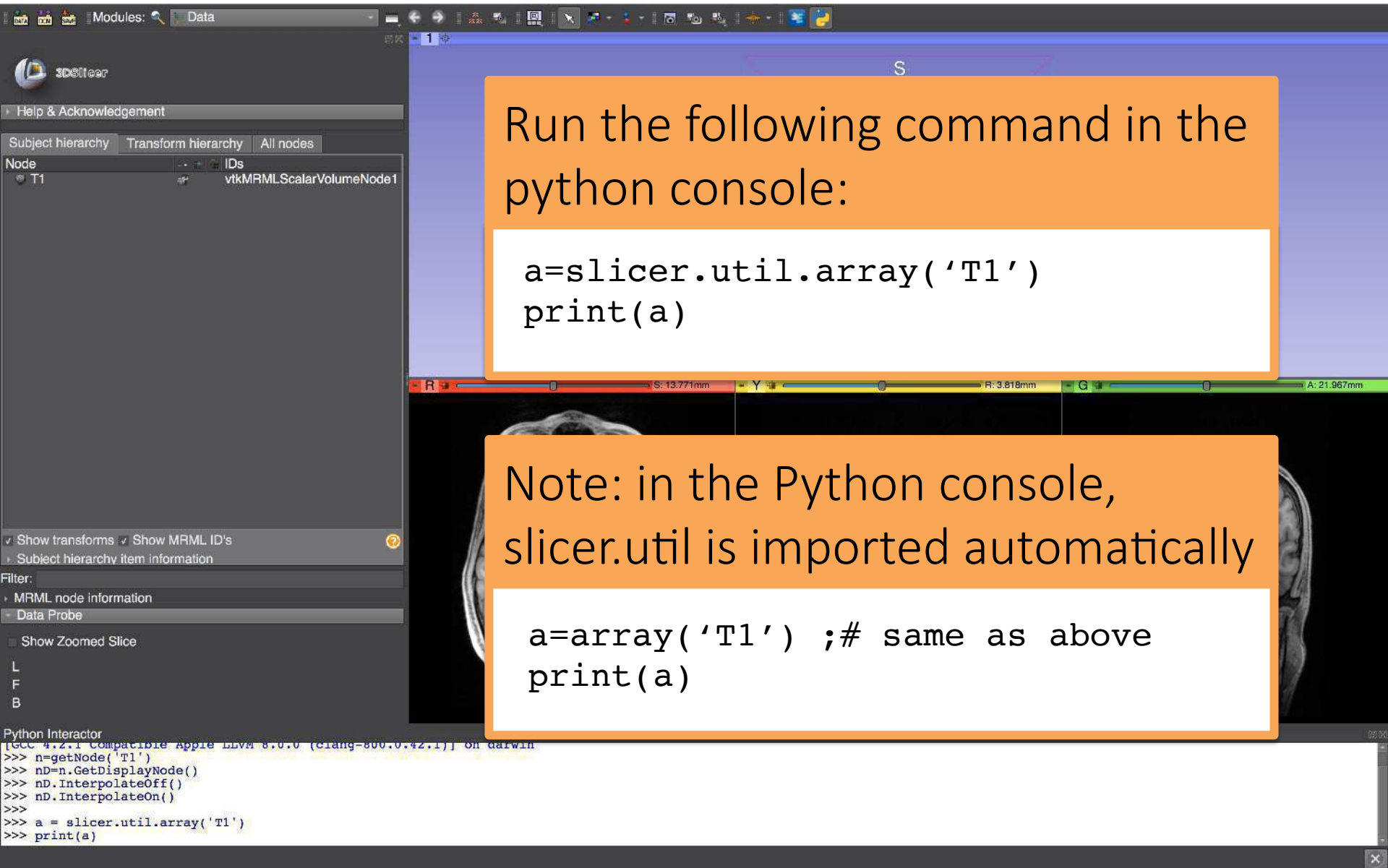
The 'Show Zoomed Slice' checkbox is checked.

Accessing voxels in a volume

- The `slicer.util` package gives access to volumes as NumPy multidimensional **arrays**
- Volumes can be modified using standard NumPy methods



Accessing voxels in a volume



The screenshot shows the Slicer 3.12.0 interface. On the left, the 'Subject hierarchy' panel shows a node 'T1' of type 'vtkMRMLScalarVolumeNode1'. Below it, the 'Filter' panel shows 'MRML node information' and 'Data Probe'. At the bottom, the 'Python Interactor' console shows the following code:

```
[GCC 4.2.1 Compatible Apple LLVM 8.0.0 (clang-800.0.42.1)] on darwin
>>> n=getNode('T1')
>>> nd=n.GetDisplayNode()
>>> nd.InterpolateOff()
>>> nd.InterpolateOn()
>>>
>>> a = slicer.util.array('T1')
>>> print(a)
```

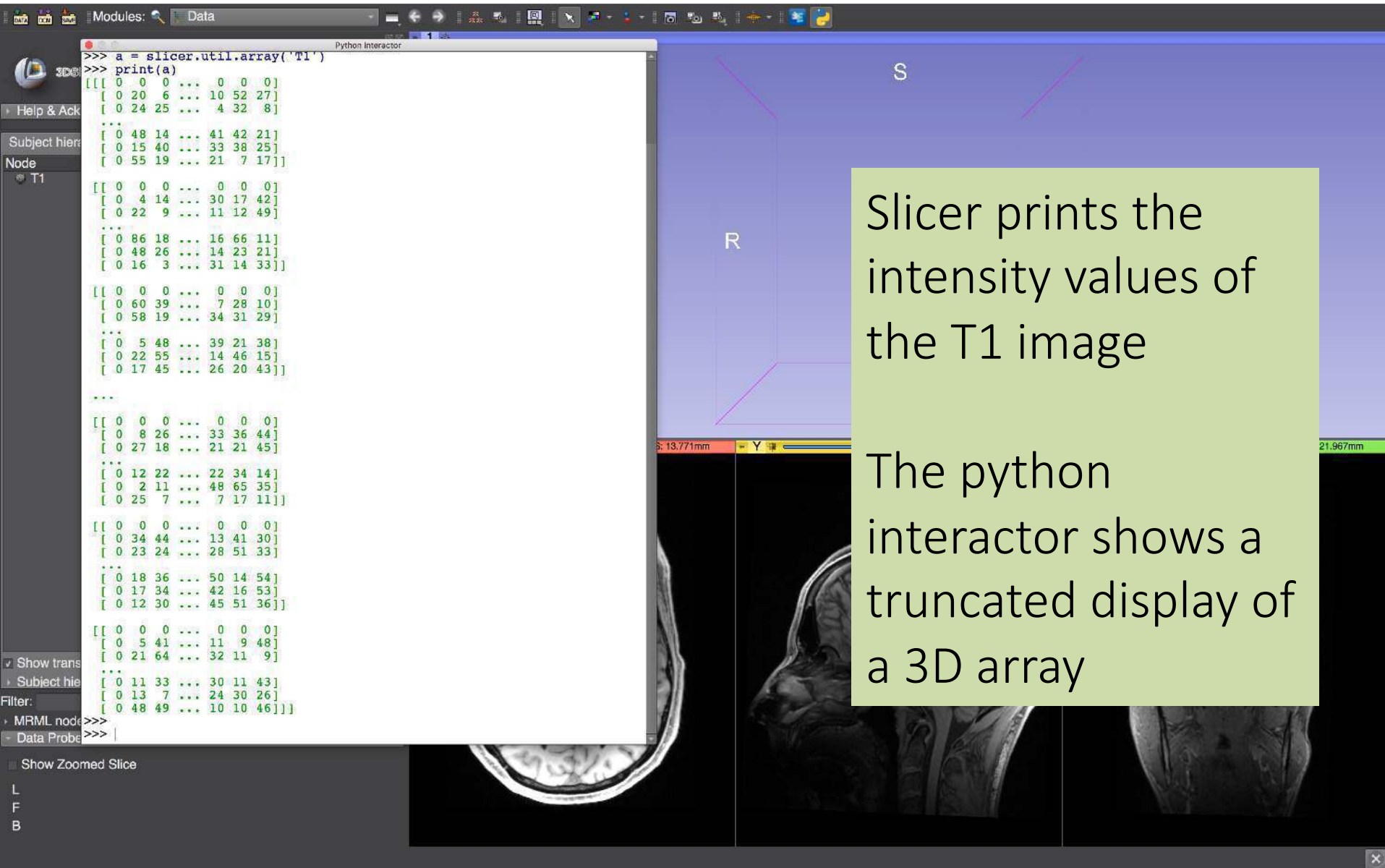
On the right, an orange box contains the text: 'Run the following command in the python console:'

```
a=slicer.util.array('T1')
print(a)
```

Below this, another orange box contains the text: 'Note: in the Python console, slicer.util is imported automatically'

```
a=array('T1') ;# same as above
print(a)
```

Accessing voxels in a volume



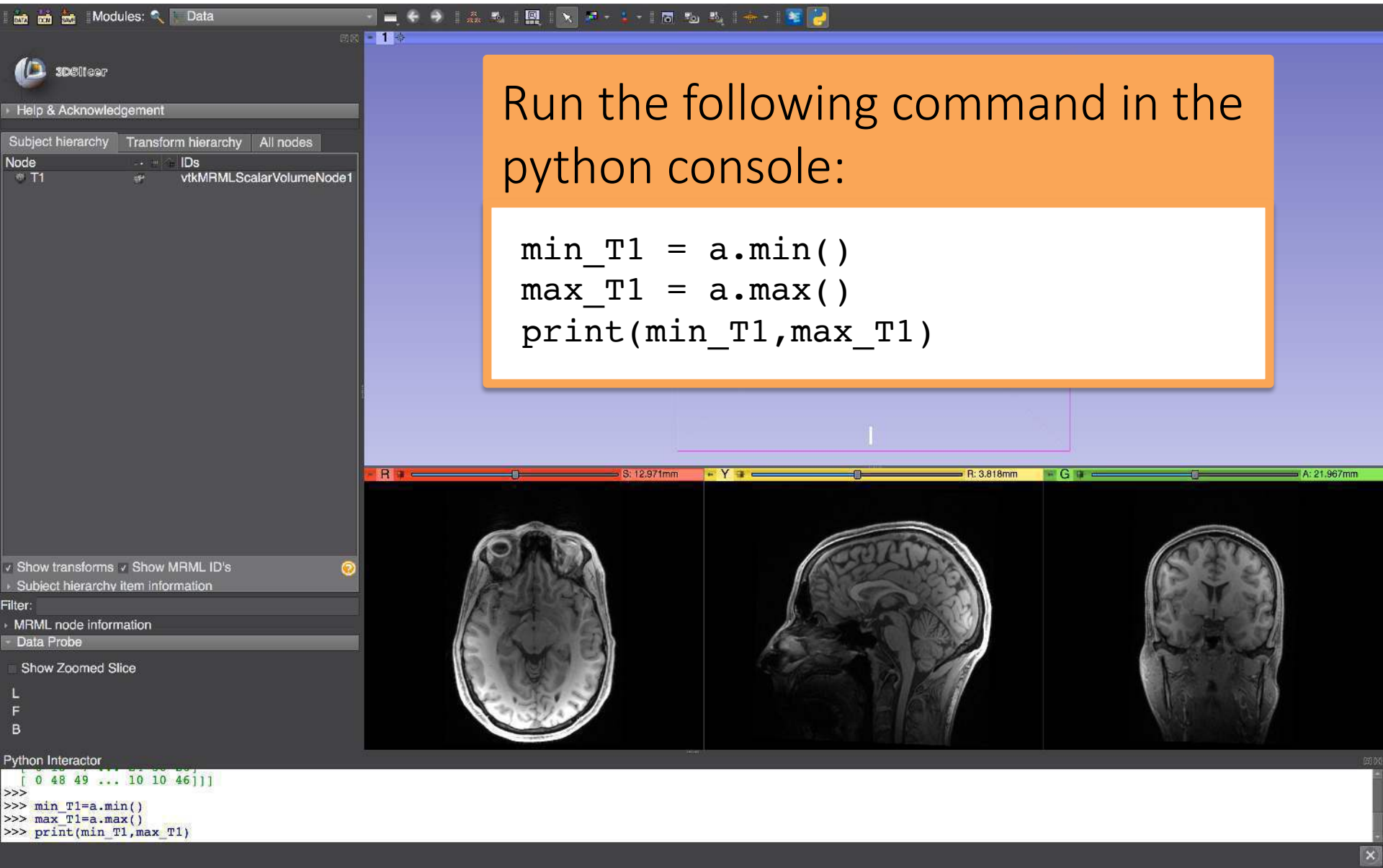
The screenshot shows the Slicer software interface. On the left, a Python Interactor window displays a truncated 3D array of intensity values for a T1 image. The array is shown in a truncated format, with ellipses indicating that the data continues beyond what is visible. The array is organized into several groups of three rows each, representing different slices or regions of the volume. The values are integers ranging from 0 to 64. On the right, a 3D visualization of the brain volume is shown, with a green box highlighting a specific region of interest. The box is labeled 'R' and 'S', indicating its position in the 3D space. The background of the 3D view is a light blue color.

```
>>> a = slicer.util.array('T1')
>>> print(a)
[[[ 0  0  0 ...  0  0  0]
 [ 0 20  6 ... 10 52 27]
 [ 0 24 25 ...  4 32  8]
 ...
 [ 0 48 14 ... 41 42 21]
 [ 0 15 40 ... 33 38 25]
 [ 0 55 19 ... 21  7 17]]
...
[[[ 0  0  0 ...  0  0  0]
 [ 0  4 14 ... 30 17 42]
 [ 0 22  9 ... 11 12 49]
 ...
 [ 0 86 18 ... 16 66 11]
 [ 0 48 26 ... 14 23 21]
 [ 0 16  3 ... 31 14 33]]
...
[[[ 0  0  0 ...  0  0  0]
 [ 0 60 39 ...  7 28 10]
 [ 0 58 19 ... 34 31 29]
 ...
 [ 0  5 48 ... 39 21 38]
 [ 0 22 55 ... 14 46 15]
 [ 0 17 45 ... 26 20 43]]
...
[[[ 0  0  0 ...  0  0  0]
 [ 0  8 26 ... 33 36 44]
 [ 0 27 18 ... 21 21 45]
 ...
 [ 0 12 22 ... 22 34 14]
 [ 0  2 11 ... 48 65 35]
 [ 0 25  7 ...  7 17 11]]
...
[[[ 0  0  0 ...  0  0  0]
 [ 0 34 44 ... 13 41 30]
 [ 0 23 24 ... 28 51 33]
 ...
 [ 0 18 36 ... 50 14 54]
 [ 0 17 34 ... 42 16 53]
 [ 0 12 30 ... 45 51 36]]
...
[[[ 0  0  0 ...  0  0  0]
 [ 0  5 41 ... 11  9 48]
 [ 0 21 64 ... 32 11  9]
 ...
 [ 0 11 33 ... 30 11 43]
 [ 0 13  7 ... 24 30 26]
 [ 0 48 49 ... 10 10 46]]]
```

Slicer prints the intensity values of the T1 image

The python interactor shows a truncated display of a 3D array

Accessing voxels in a volume

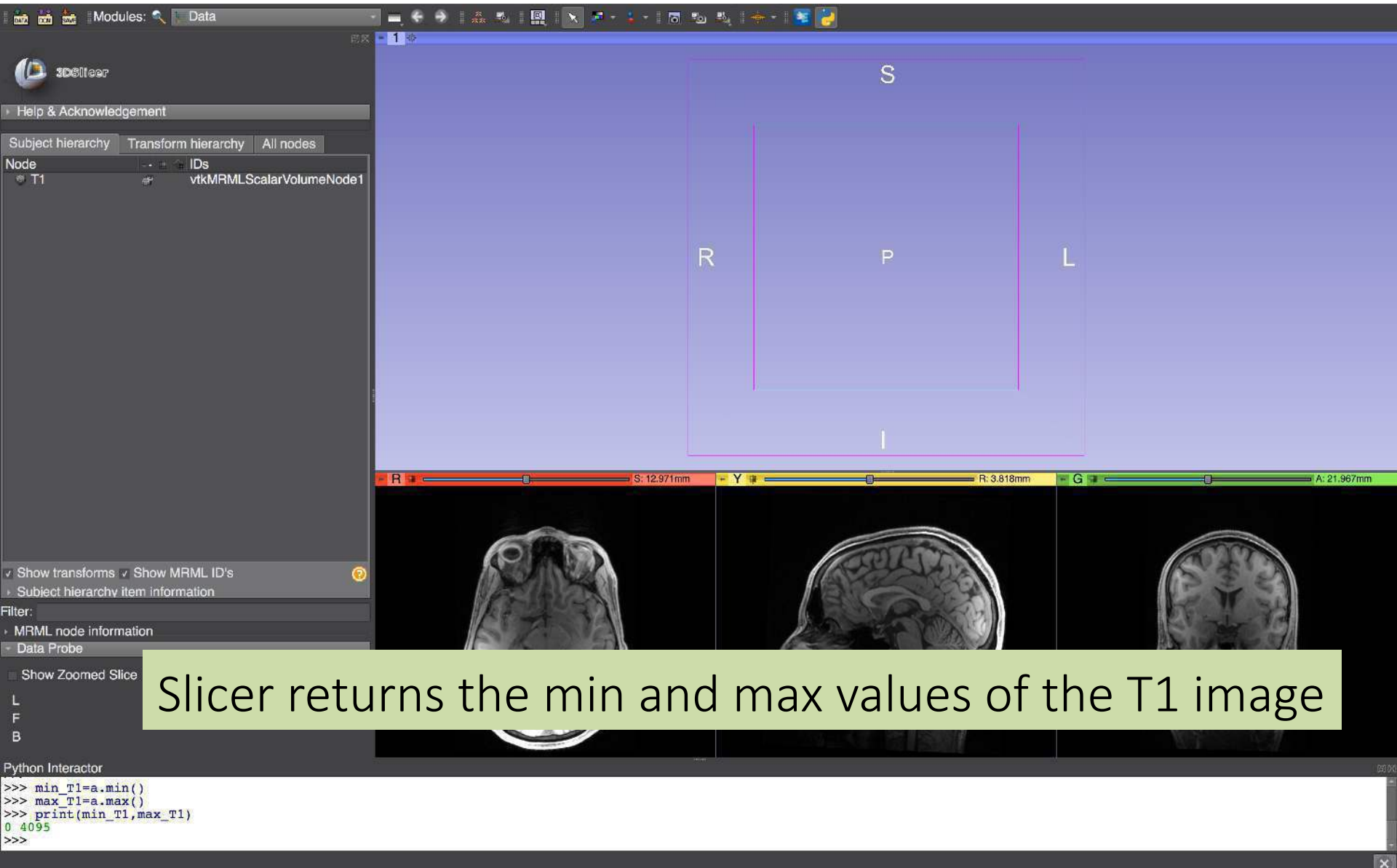


Run the following command in the python console:

```
min_T1 = a.min()  
max_T1 = a.max()  
print(min_T1,max_T1)
```

The screenshot shows the 3D Slicer software interface. The left sidebar contains the 'Subject hierarchy' panel with a tree view showing 'T1' and 'vtkMRMLScalarVolumeNode1'. Below it are checkboxes for 'Show transforms', 'Show MRML ID's', and 'Subject hierarchy item information'. The bottom left corner features a 'Python Interactor' window with a list of data points and the execution of the commands shown in the orange box. The main window displays three orthogonal views of a brain MRI volume: axial, sagittal, and coronal. Above the views are sliders for 'R' (12.971mm), 'Y' (3.818mm), and 'G' (21.967mm).

Accessing voxels in a volume

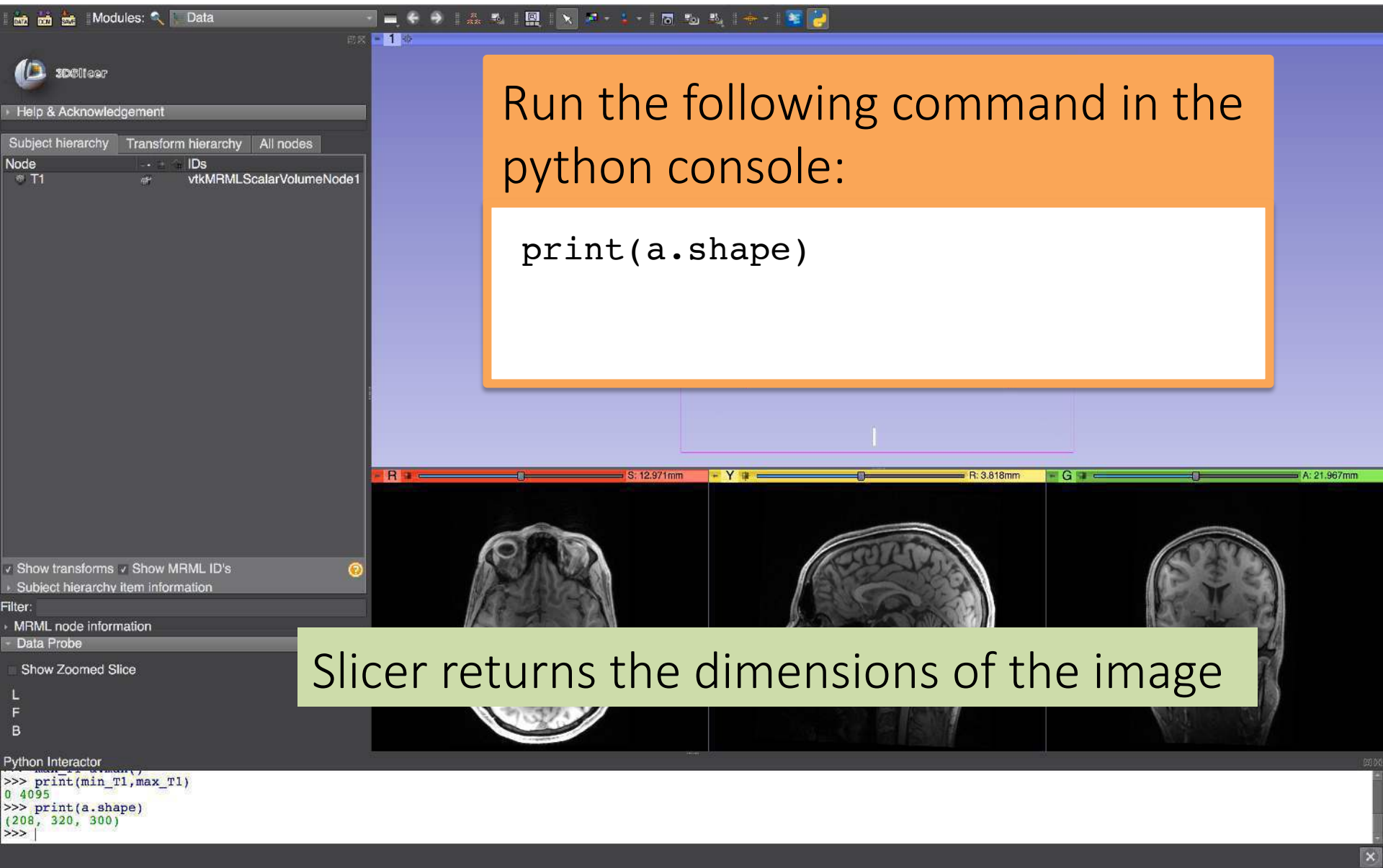


The screenshot displays the 3D Slicer software interface. The main window shows a T1 brain volume with a purple bounding box. The bounding box is labeled with 'S' at the top, 'I' at the bottom, 'R' on the left, and 'L' on the right. Below the main window, there are three orthogonal views: axial, sagittal, and coronal. The axial view is on the left, the sagittal view is in the middle, and the coronal view is on the right. Above the views, there are sliders for the R, Y, and G planes, with values S: 12.971mm, R: 3.818mm, and A: 21.967mm respectively. On the left side, there is a sidebar with tabs for 'Subject hierarchy', 'Transform hierarchy', and 'All nodes'. The 'Subject hierarchy' tab is active, showing a tree structure with 'Node' and 'IDs' columns. The 'Node' column shows 'T1' and the 'IDs' column shows 'vtkMRMLScalarVolumeNode1'. Below the sidebar, there is a 'Filter' section with 'MRML node information' and 'Data Probe' options. At the bottom, there is a 'Python Interactor' window with the following code and output:

```
>>> min_T1=a.min()
>>> max_T1=a.max()
>>> print(min_T1,max_T1)
0 4095
>>>
```

Slicer returns the min and max values of the T1 image

Modifying voxels in a volume



The screenshot shows the 3D Slicer software interface. On the left is a sidebar with a 'Subject hierarchy' panel showing a tree structure with 'Node T1' and 'vtkMRMLScalarVolumeNode1'. Below it are checkboxes for 'Show transforms', 'Show MRML ID's', and 'Subject hierarchy item information'. At the bottom left is a 'Python Interactor' console. The main window displays three orthogonal views of a brain MRI volume: axial, sagittal, and coronal. Above the views are sliders for 'R' (12.971mm), 'Y' (3.818mm), and 'G' (21.967mm). An orange text box is overlaid on the main window, and a green text box is overlaid at the bottom.

Run the following command in the python console:

```
print(a.shape)
```

Slicer returns the dimensions of the image

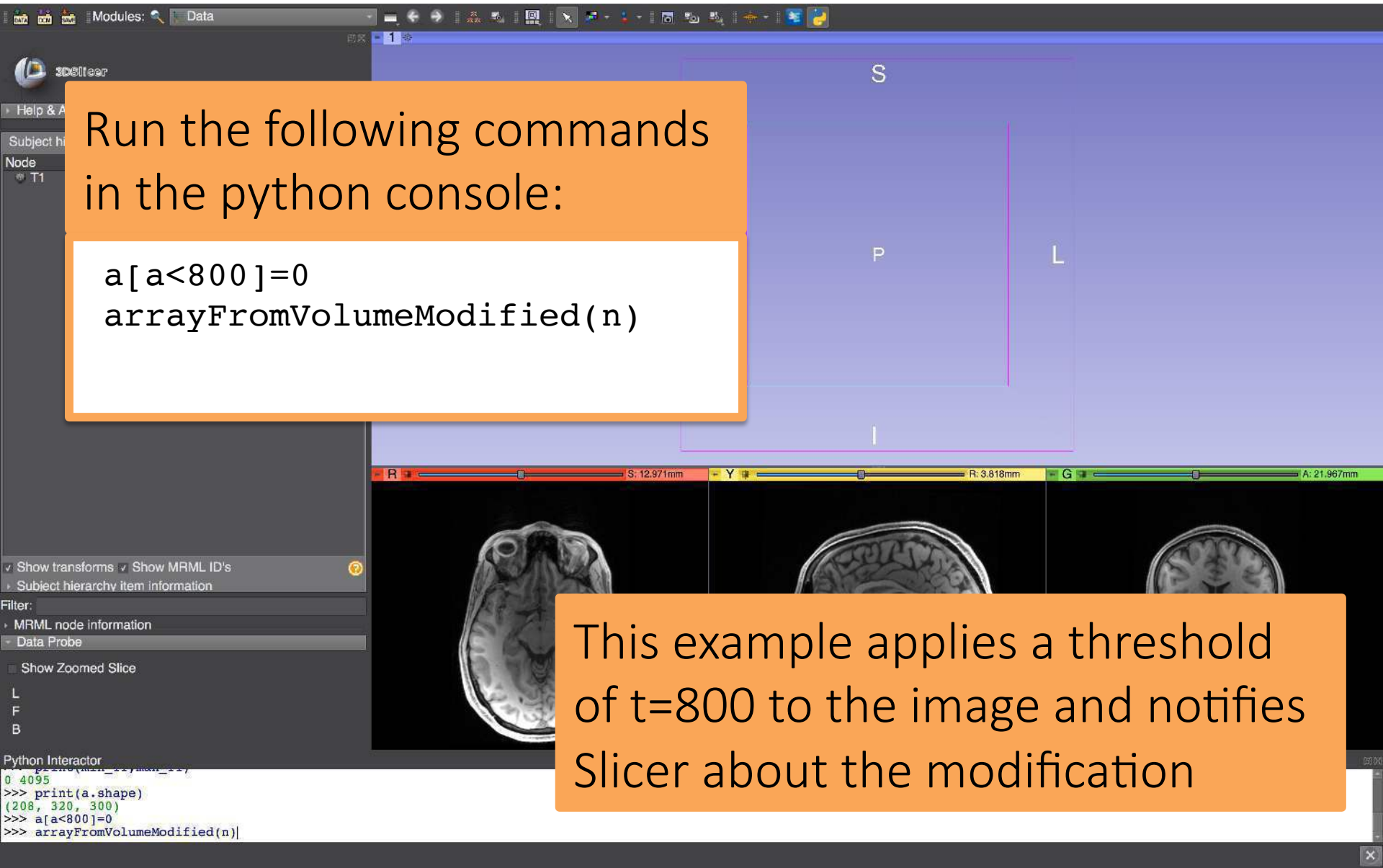
```
>>> print(min_T1,max_T1)
0 4095
>>> print(a.shape)
(208, 320, 300)
>>>
```

Modifying voxels in a volume

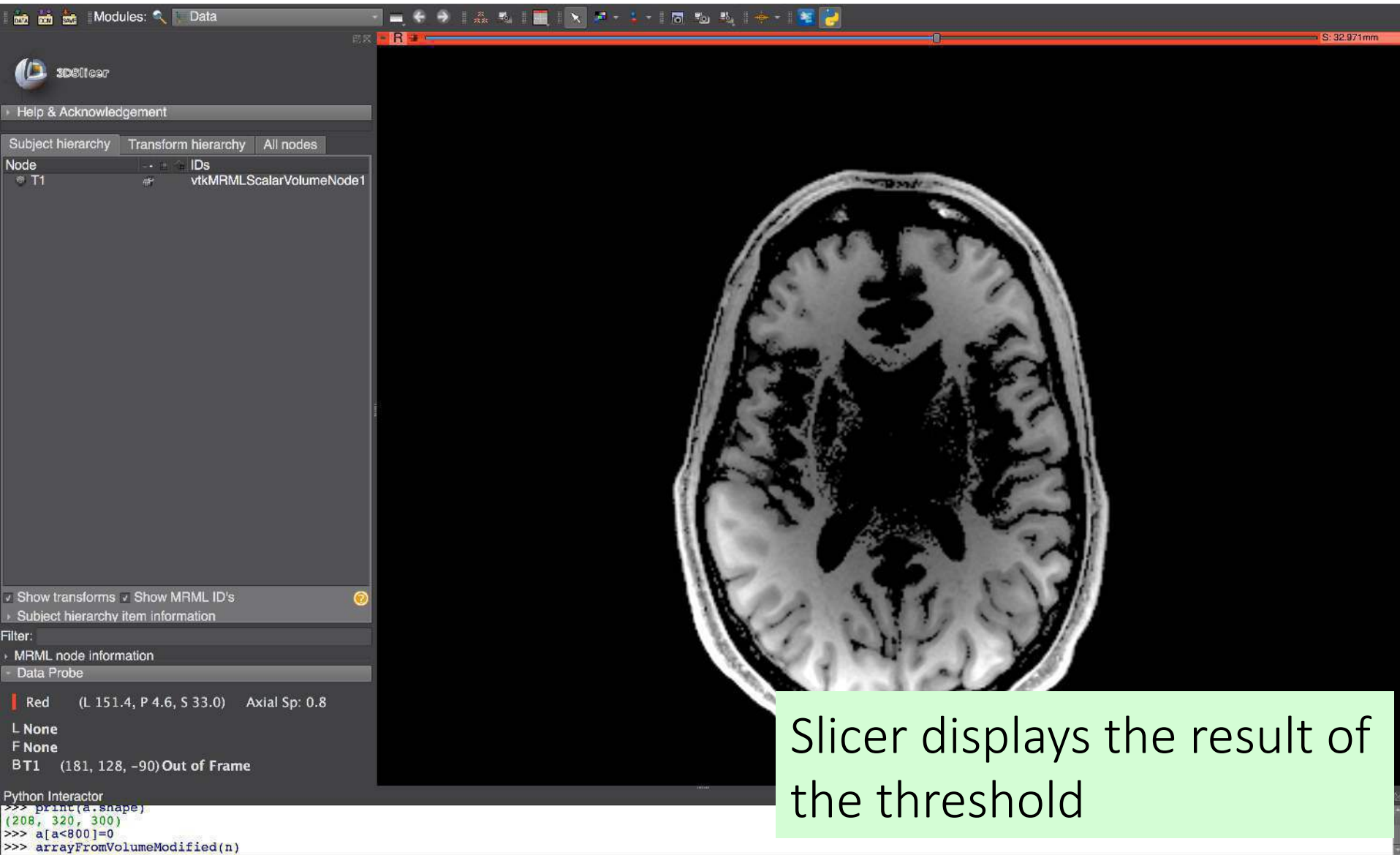
Run the following commands in the python console:

```
a[a<800]=0  
arrayFromVolumeModified(n)
```

This example applies a threshold of $t=800$ to the image and notifies Slicer about the modification



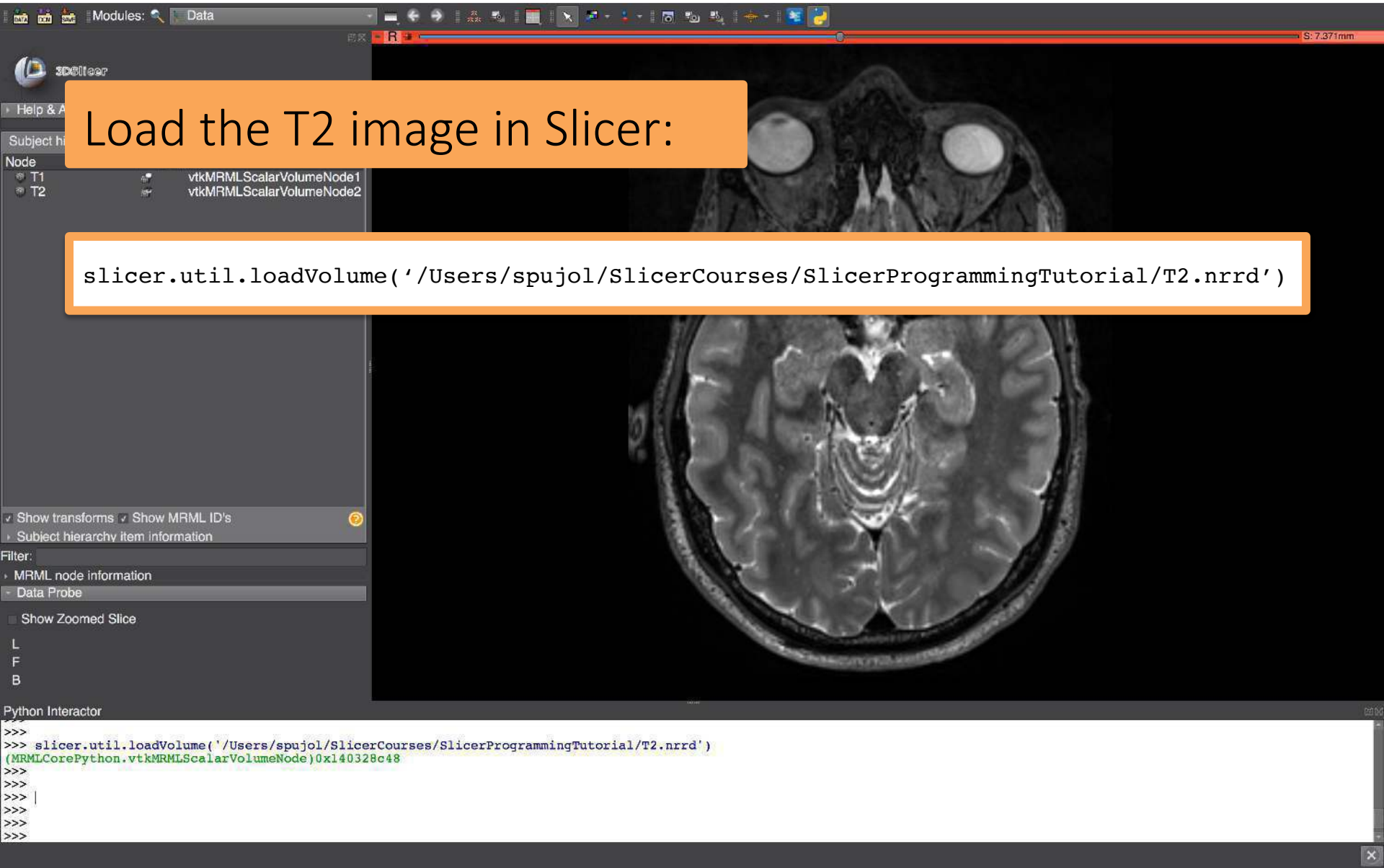
Modifying voxels in a volume



Loading the T2 volume

Load the T2 image in Slicer:

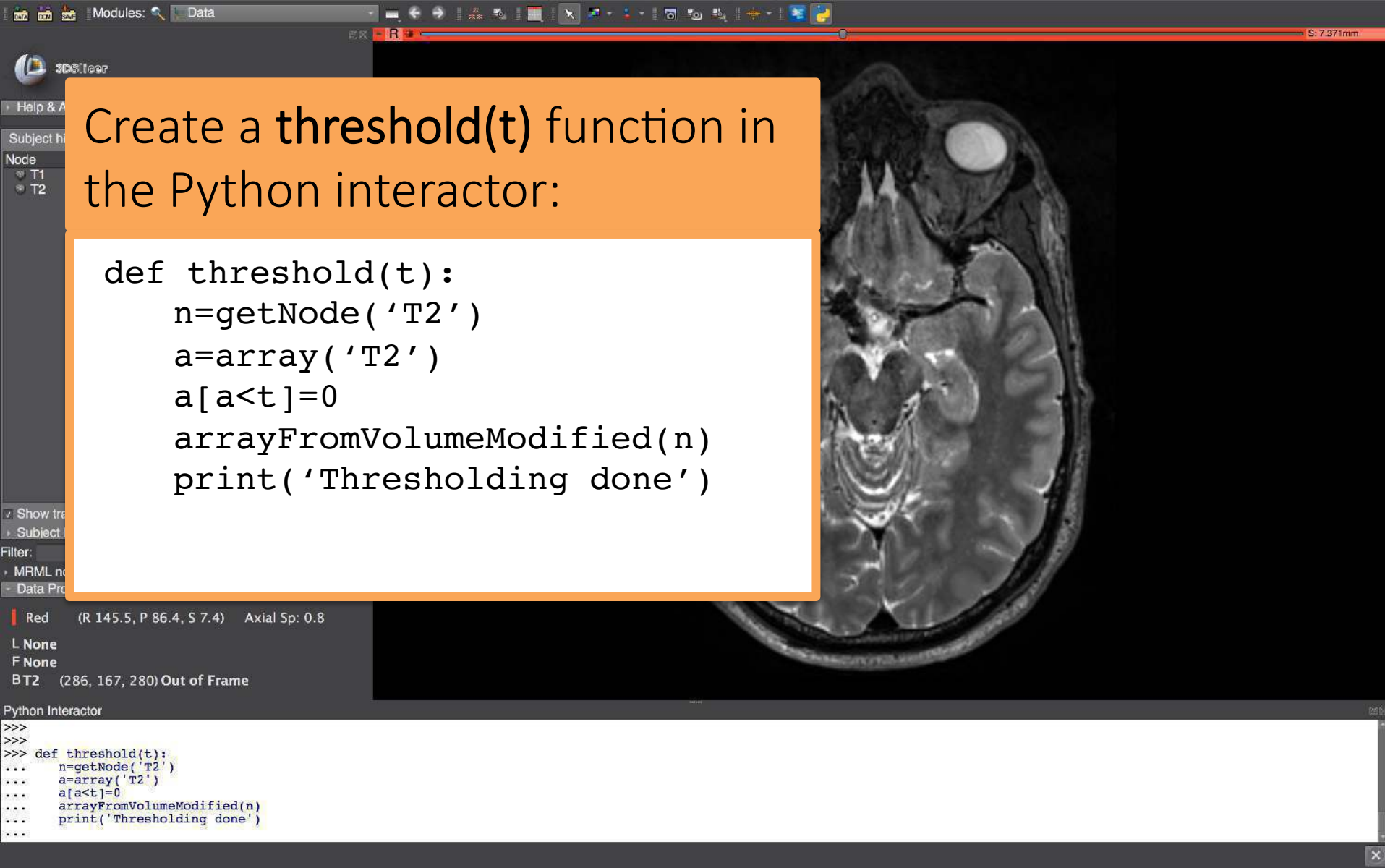
```
slicer.util.loadVolume('/Users/spujol/SlicerCourses/SlicerProgrammingTutorial/T2.nrrd')
```



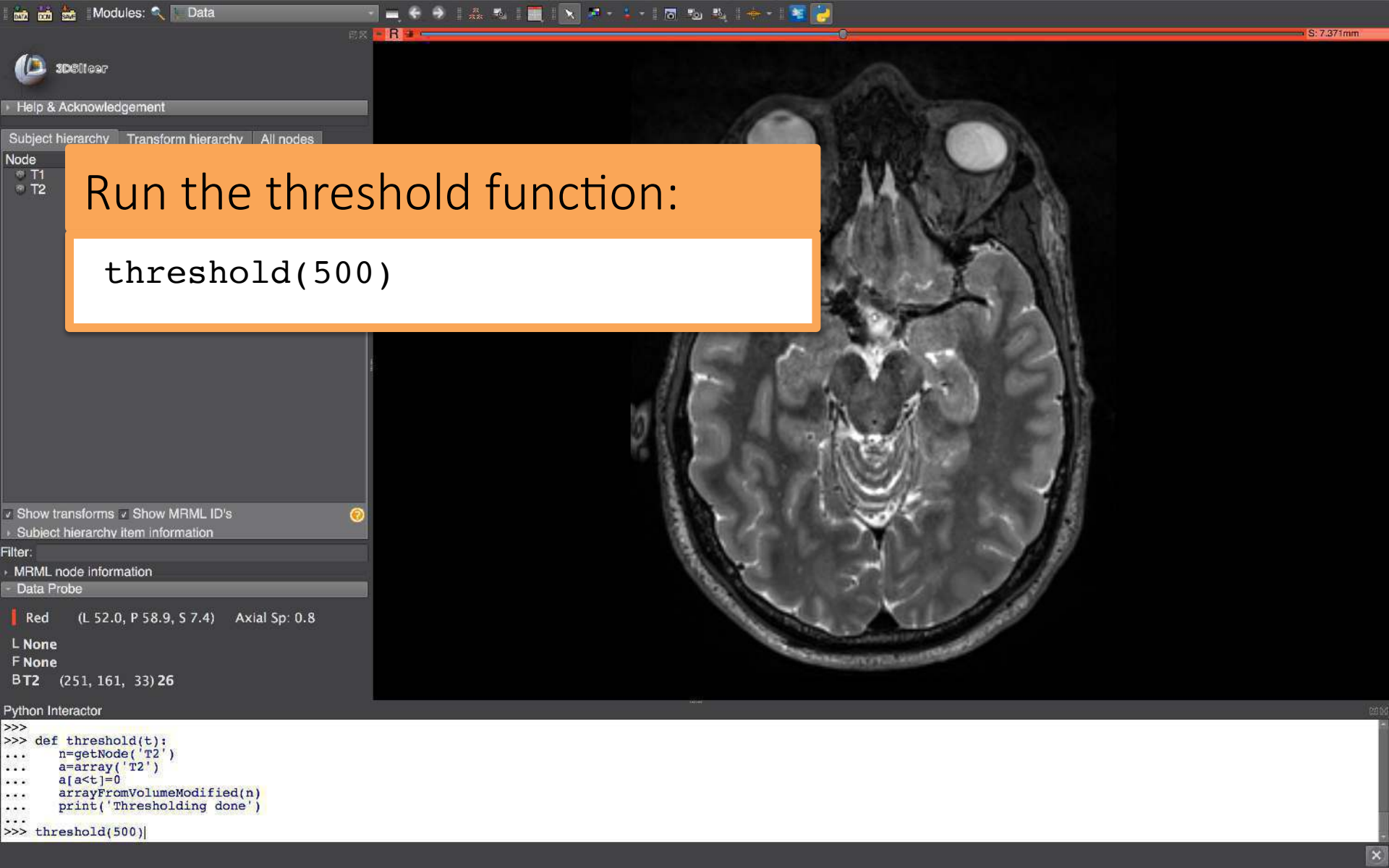
Python function: threshold

Create a **threshold(t)** function in the Python interactor:

```
def threshold(t):  
    n=getNode('T2')  
    a=array('T2')  
    a[a<t]=0  
    arrayFromVolumeModified(n)  
    print('Thresholding done')
```



Python function: threshold



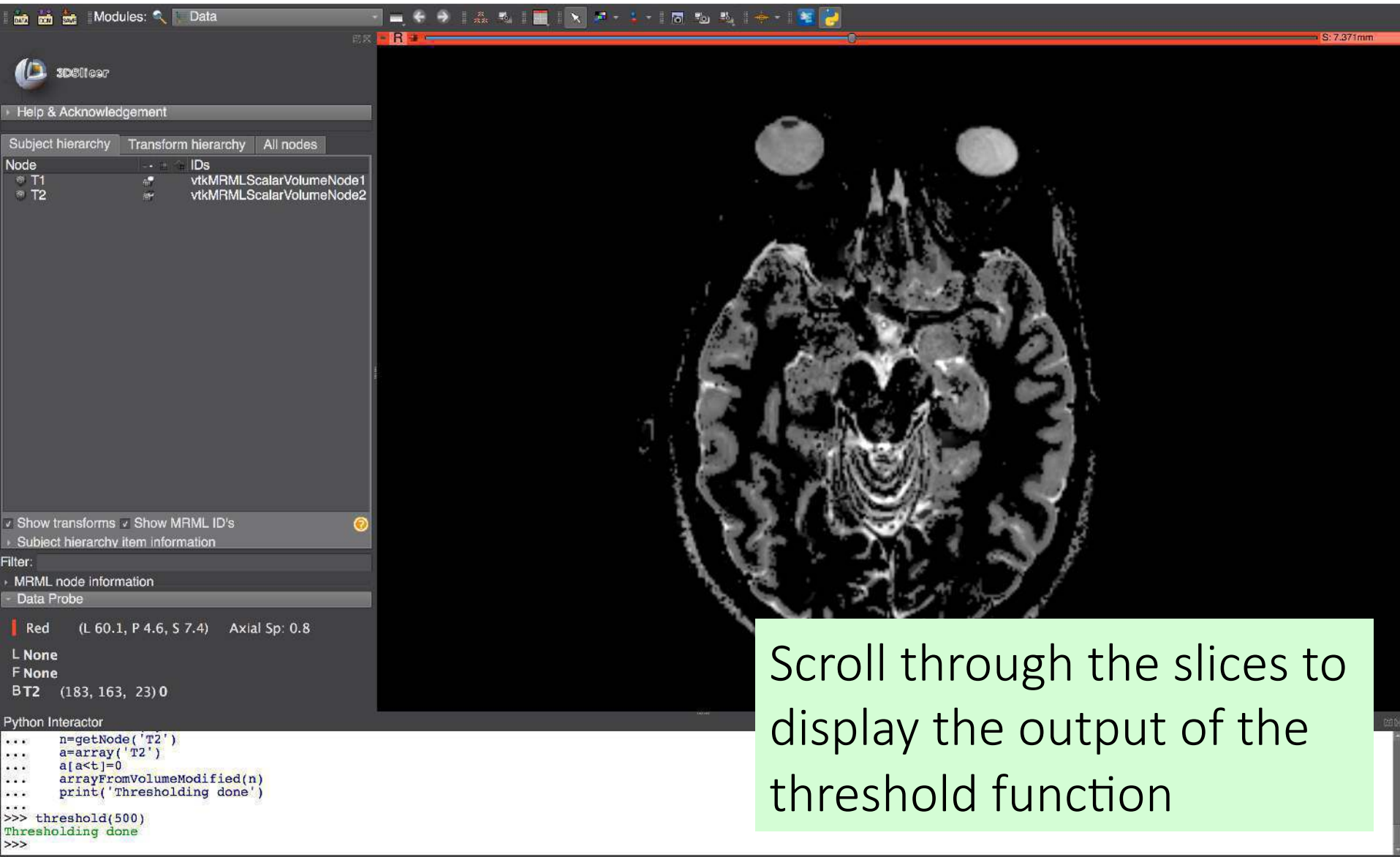
Run the threshold function:

```
threshold(500)
```

Python Interactor

```
>>>
>>> def threshold(t):
...     n=getNode('T2')
...     a=array('T2')
...     a[a<t]=0
...     arrayFromVolumeModified(n)
...     print('Thresholding done')
...
>>> threshold(500)
```


Python function: threshold



The screenshot shows the 3D Slicer software interface. The left sidebar contains a hierarchy panel with nodes T1 and T2, a filter panel with options for MRML node information and Data Probe, and a Python Interactor. The main window displays an axial brain MRI slice. A green text box on the right contains the following text:

Scroll through the slices to display the output of the threshold function

```
Python Interactor
... n=getNode('T2')
... a=array('T2')
... a[a<t]=0
... arrayFromVolumeModified(n)
... print('Thresholding done')
...
>>> threshold(500)
Thresholding done
>>>
```

Big Picture

- Slicer provides easy access to analyze and modify complex data types
- Slicer is compatible with a wide range of Python scientific computing packages
- Slicer is a research environment for performing medical imaging experiments

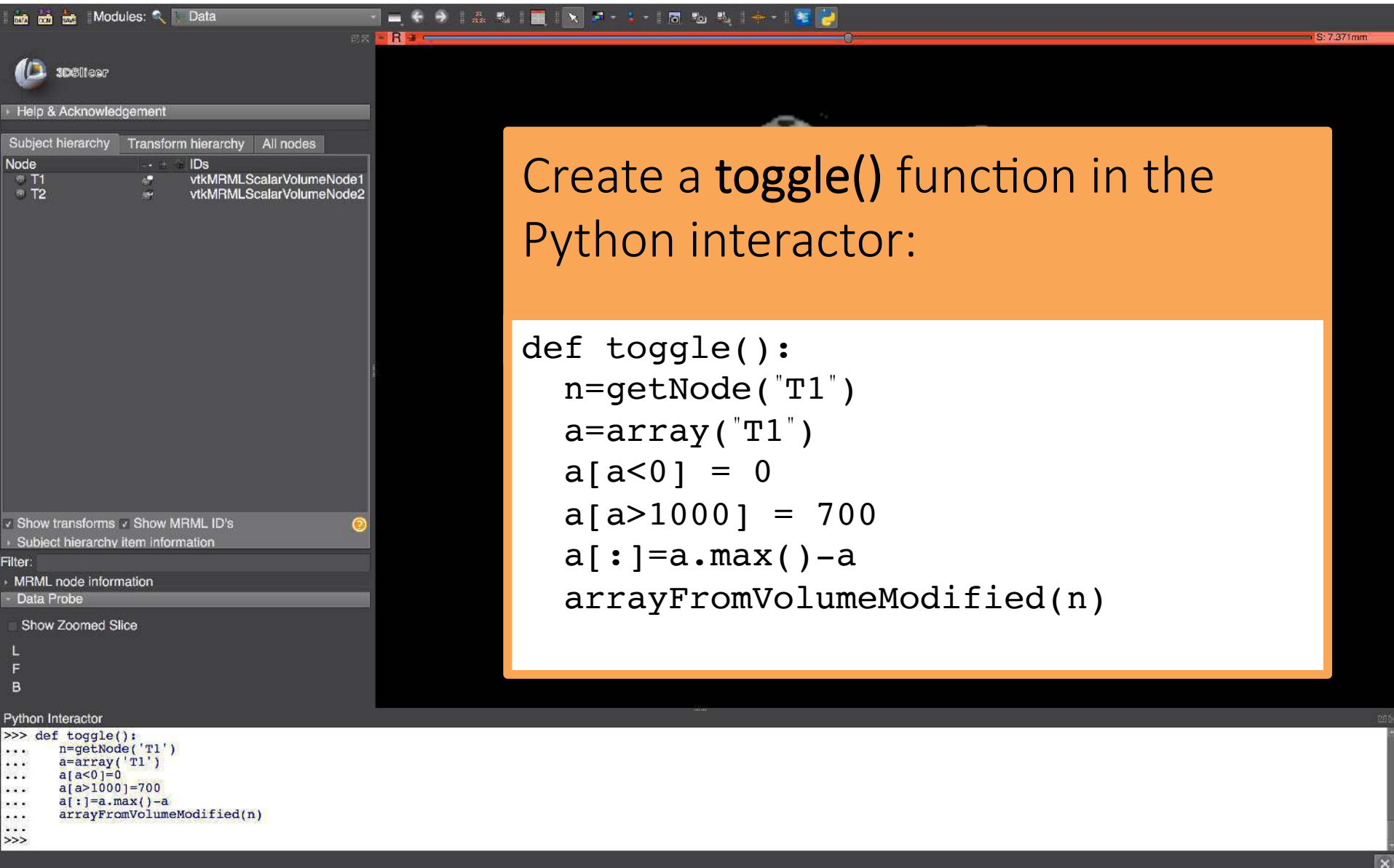
Part 3

Getting familiar with Qt in Slicer

Qt & PythonQt

- **Qt** is the main tool in Slicer to create widgets, dialogs, text entries, etc.
- **PythonQt** exposes most Qt functionalities and is accessible through the Python interactor in Slicer
- User interfaces can be created on the fly for rapid prototyping and debugging

Python function: toggle

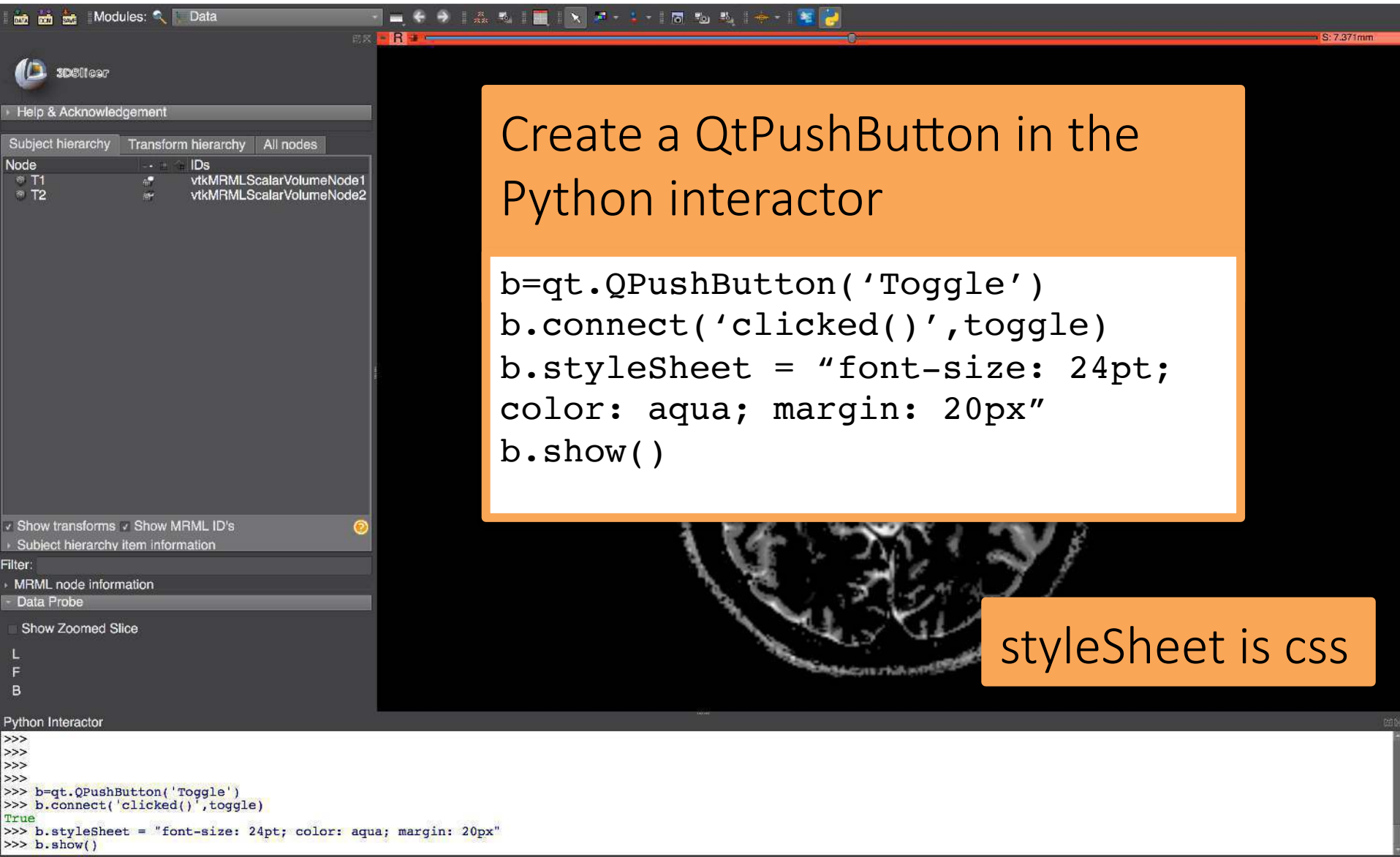


The screenshot shows the 3D Slicer software interface. On the left, the 'Node' list is visible, showing two nodes: 'T1' (vtkMRMLScalarVolumeNode1) and 'T2' (vtkMRMLScalarVolumeNode2). Below the node list, there are checkboxes for 'Show transforms', 'Show MRML ID's', and 'Subject hierarchy item information'. At the bottom left, the 'Python Interactor' window is open, displaying the following code:

```
>>> def toggle():
...     n=getNode('T1')
...     a=array('T1')
...     a[a<0]=0
...     a[a>1000]=700
...     a[:]=a.max()-a
...     arrayFromVolumeModified(n)
...
>>>
```

On the right, an orange box contains the text: 'Create a toggle() function in the Python interactor:'.

Creating a Qt Push Button



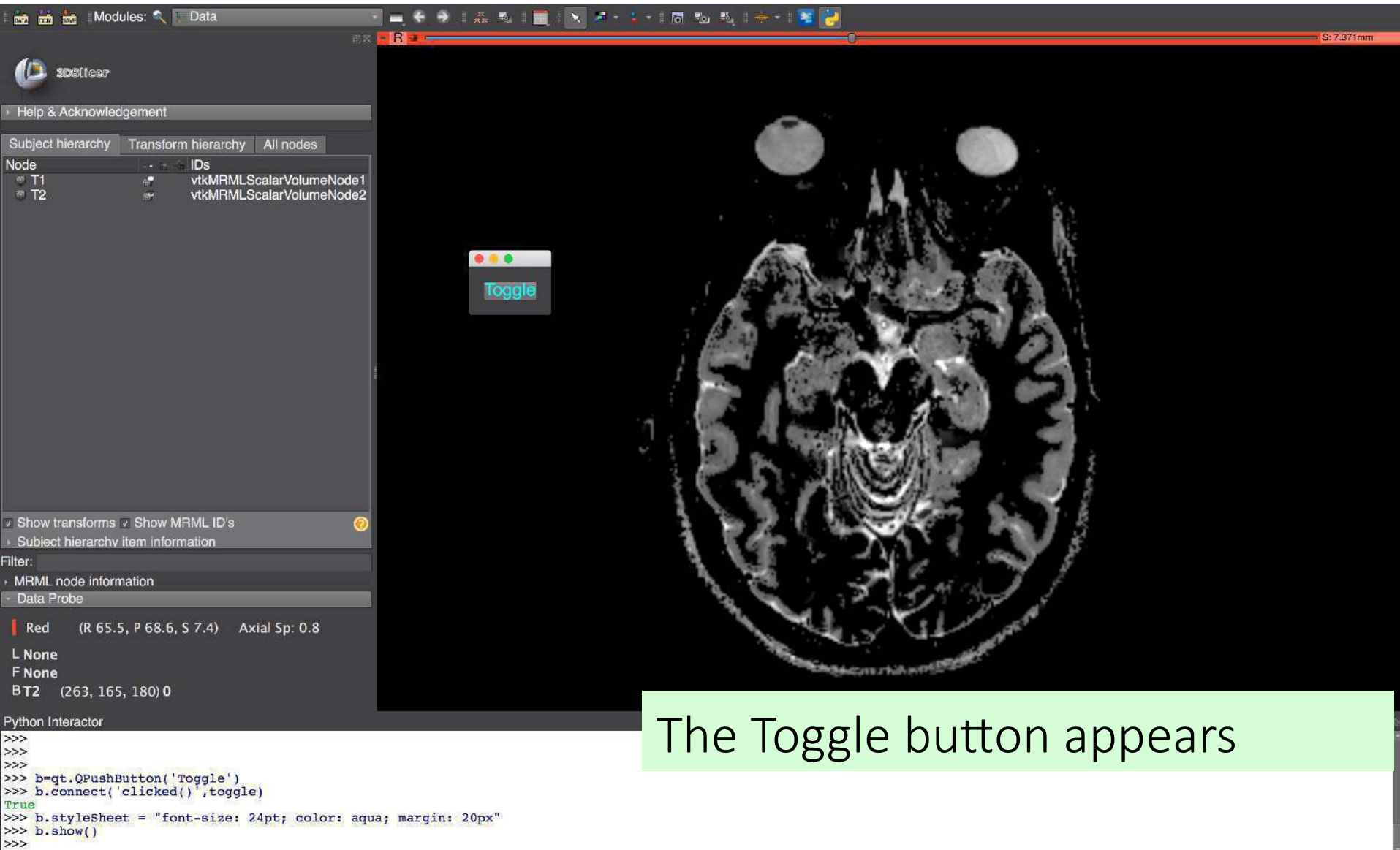
The screenshot shows the 3D Slicer software interface. On the left, the 'Subject hierarchy' panel lists two nodes: T1 (vtkMRMLScalarVolumeNode1) and T2 (vtkMRMLScalarVolumeNode2). Below this, there are checkboxes for 'Show transforms', 'Show MRML ID's', and 'Subject hierarchy item information'. The 'Filter' section shows 'MRML node information' and 'Data Probe'. At the bottom, the 'Python Interactor' window displays the following code:

```
>>>
>>>
>>>
>>>
>>> b=qt.QPushButton('Toggle')
>>> b.connect('clicked()',toggle)
True
>>> b.styleSheet = "font-size: 24pt; color: aqua; margin: 20px"
>>> b.show()
```

In the center of the image, there is an orange box with the text: "Create a QPushButton in the Python interactor".

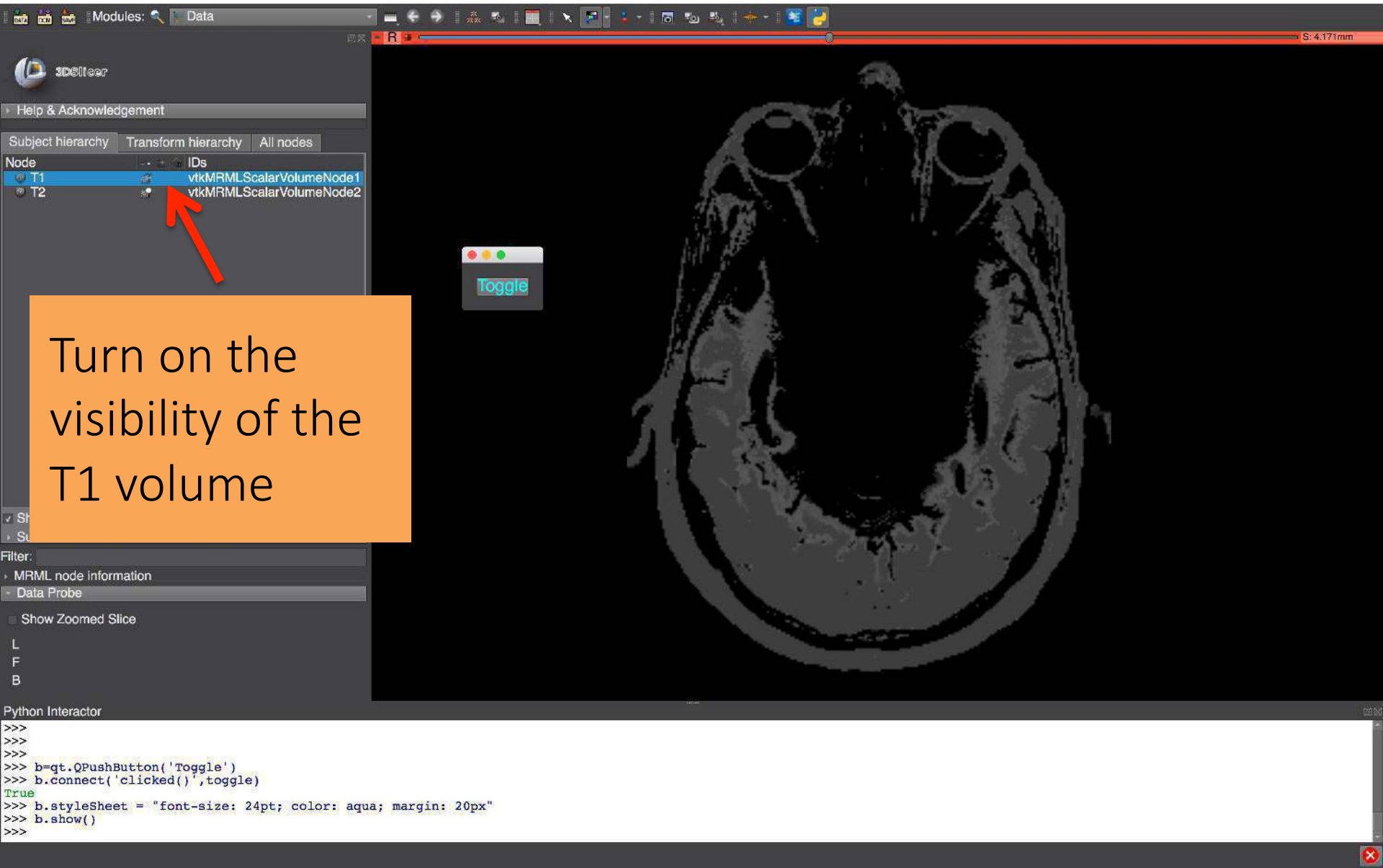
Below the code, there is another orange box with the text: "styleSheet is css".

Creating a Qt Push Button



The Toggle button appears

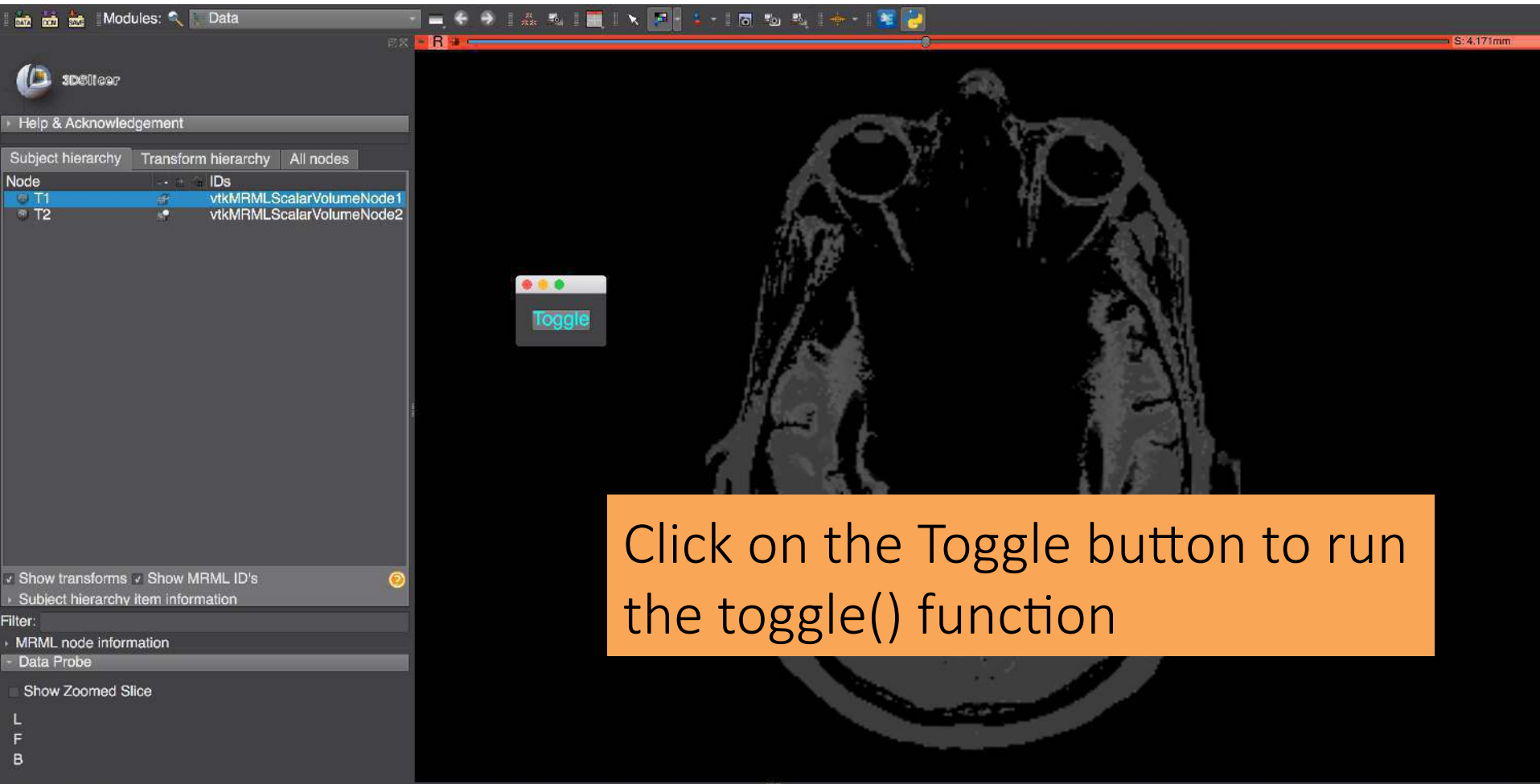
Creating a Qt Push Button



Turn on the visibility of the T1 volume

```
>>>
>>>
>>> b=qt.QPushButton('Toggle')
>>> b.connect('clicked()',toggle)
True
>>> b.setStyleSheet = "font-size: 24pt; color: aqua; margin: 20px"
>>> b.show()
>>>
```

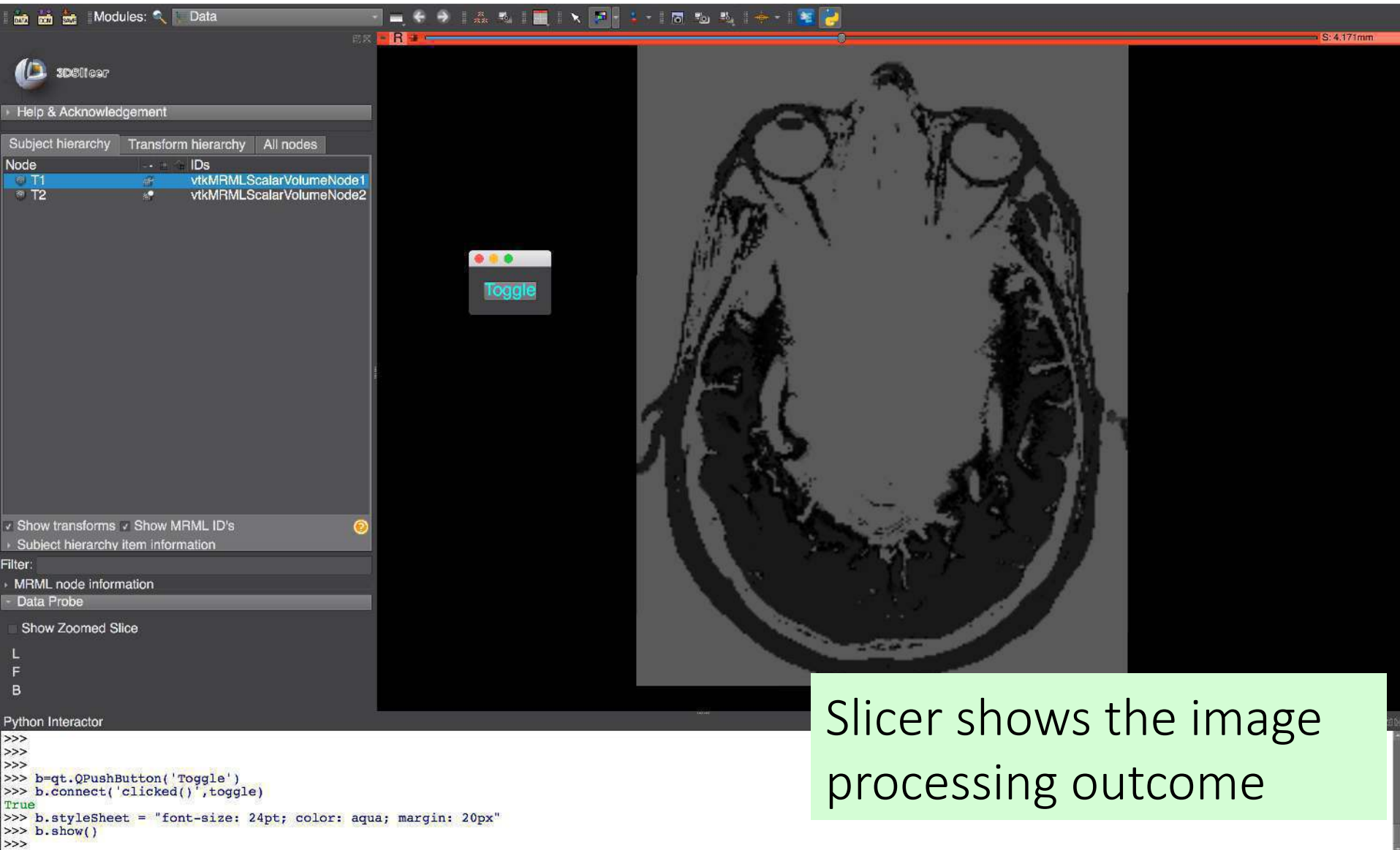

Creating a Qt Push Button



The screenshot shows the 3D Slicer software interface. On the left, the 'Subject hierarchy' panel lists two nodes: 'T1' (vtkMRMLScalarVolumeNode1) and 'T2' (vtkMRMLScalarVolumeNode2). The 'Toggle' button is visible in the center of the main 3D view area. An orange text box is overlaid on the right side of the image, containing the instruction: 'Click on the Toggle button to run the toggle() function'.

```
>>>  
>>>  
>>>  
>>> b=qt.QPushButton('Toggle')  
>>> b.connect('clicked()',toggle)  
True  
>>> b.setStyleSheet = "font-size: 24pt; color: aqua; margin: 20px"  
>>> b.show()  
>>>
```

Creating a Qt Push Button



Slicer shows the image processing outcome

Examples of scripted modules

- The tutorial demonstrates how to create a simple interface in Python
- Slicer integrates many sophisticated scripted modules such as Segment Statistics, Sample Data, Endoscopy module, etc.
- For further reading, please look at the Slicer Script Repository:

<https://www.slicer.org/wiki/Documentation/Nightly/ScriptRepository>

Conclusion

- Slicer enables developers to create complex interfaces that are streamlined for target users
- The software platform provides unlimited customization possibilities
- Slicer gives access to advanced underlying libraries through a cross-platform package that is easy to deploy to end-users

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